

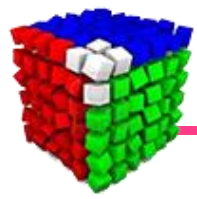
Modelos de Cores



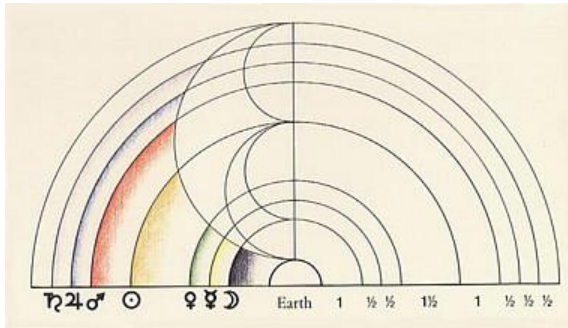
Carlos Alexandre Barros de Mello
Edward Roe



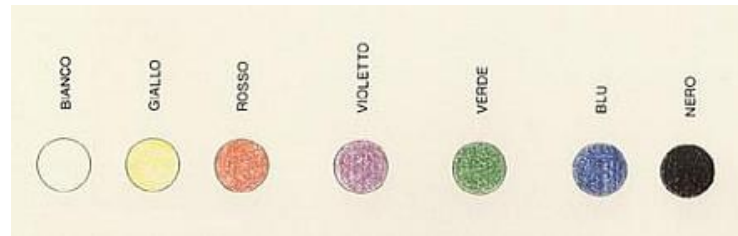
Sobre as cores....



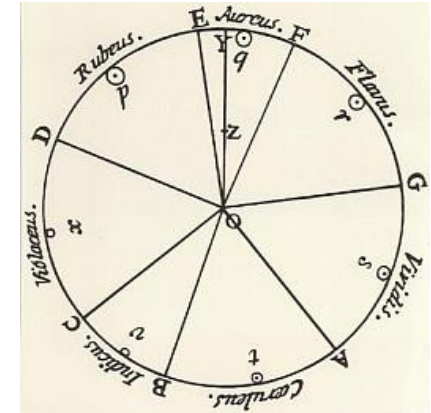
Evolução dos Modelos de Cor



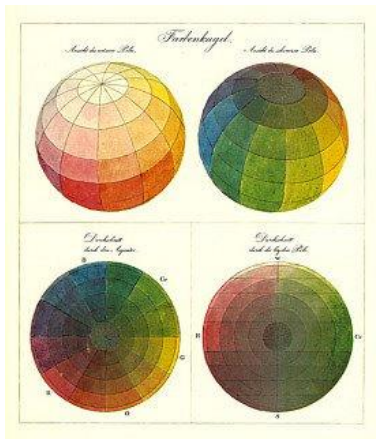
Pitágoras ~550AC



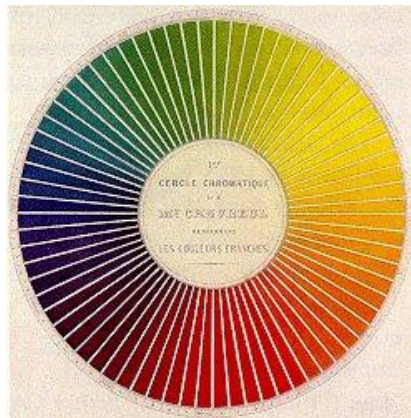
Aristóteles ~350BC



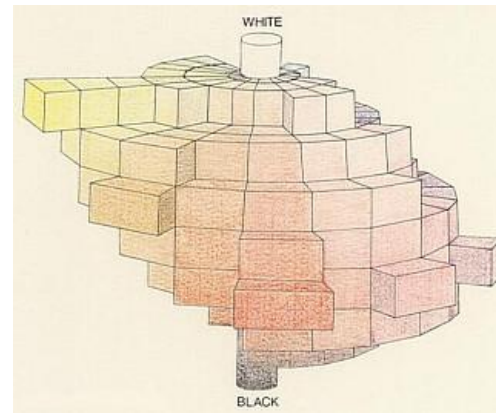
Isaac Newton - 1704



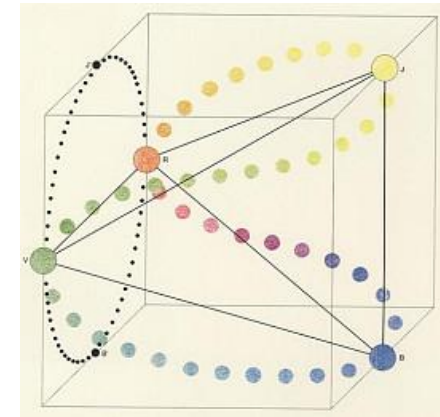
Philipp Otto Runge 1810



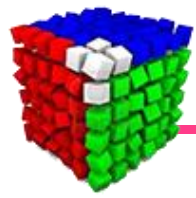
Michel Eugène Chevreul 1839



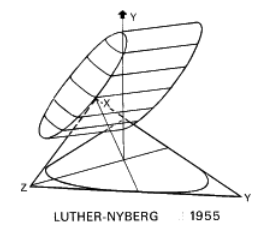
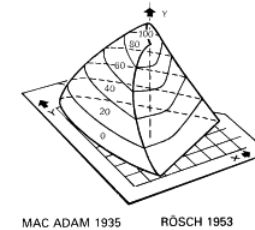
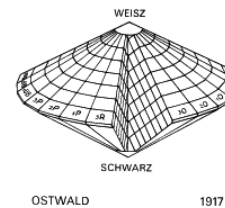
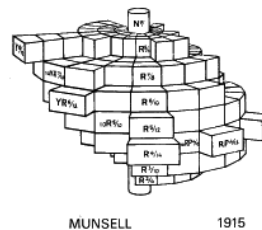
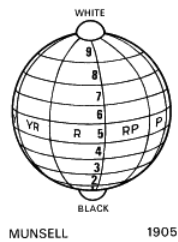
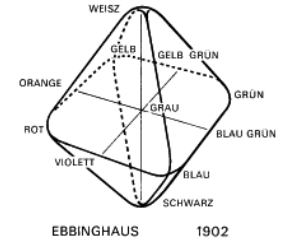
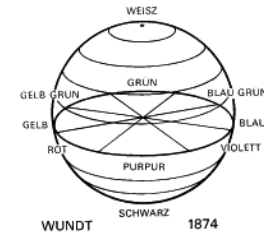
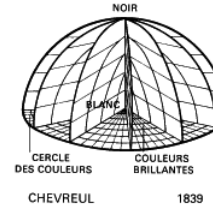
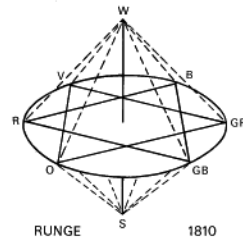
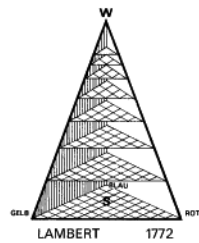
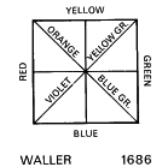
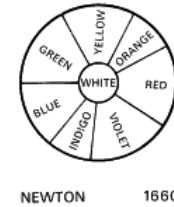
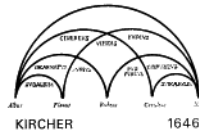
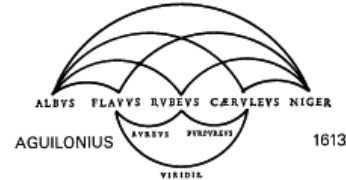
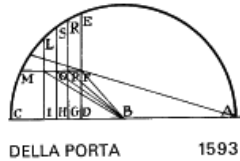
Albert Henry Munsell 1915

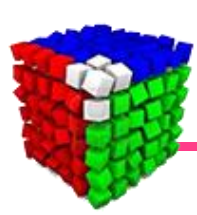


Michel Albert-Vanel 1983



Outros Modelos



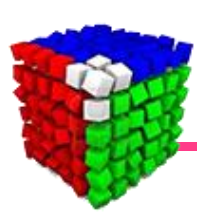


Isaac Newton

- Acreditava na teoria corpuscular da luz
- Huygens acreditava na teoria ondulatória
- Teoria de Newton foi mais aceita devido ao seu grande reconhecimento pela gravitação
- Newton fez importantes experimentos sobre a decomposição da luz com prismas e acreditou que as cores eram devidas ao tamanho da partícula de luz
- Hoje, sabe-se que a luz se comporta ora como partícula, ora como onda

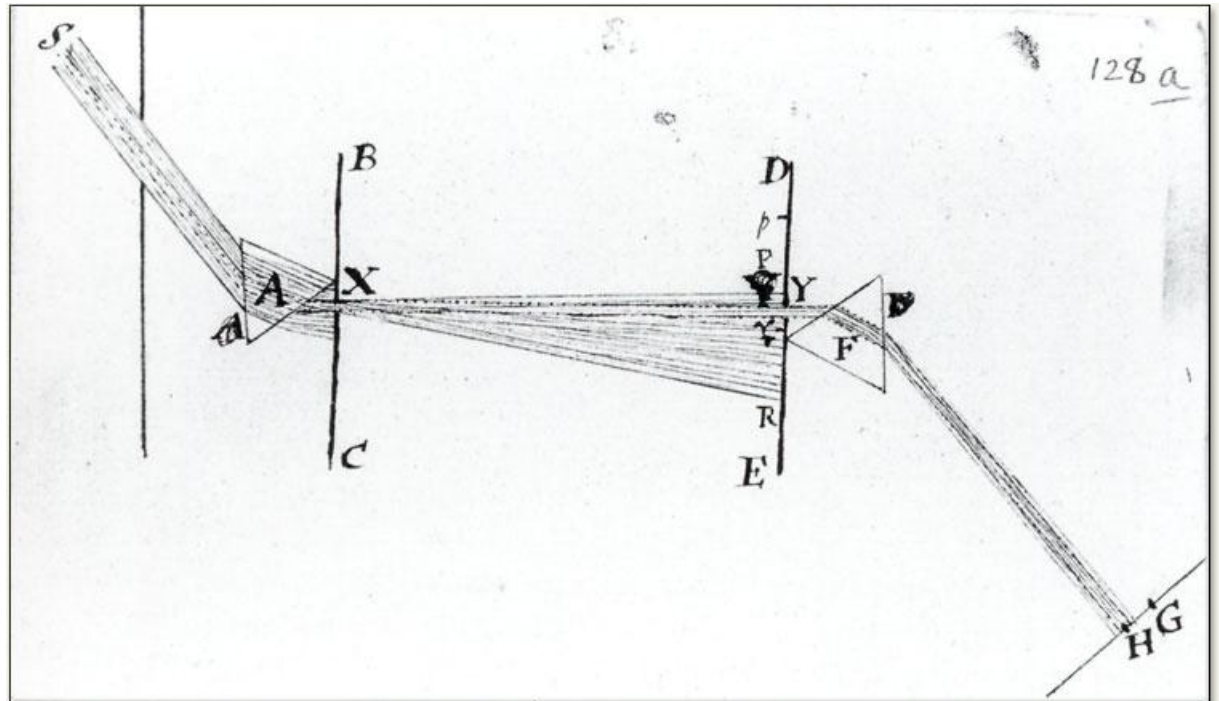
Um pouco mais sobre o assunto:

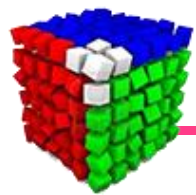
<https://www.youtube.com/watch?v=OQ6kWit51fA>



Experiência de Newton

- Através de um experimento simples Newton mostrou que a luz branca era na verdade composta por outras cores

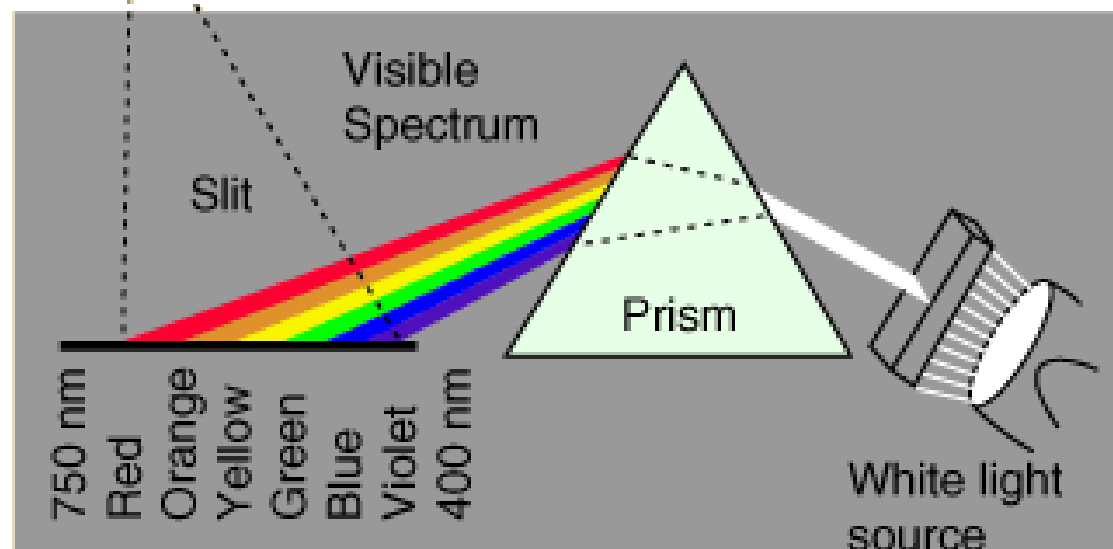


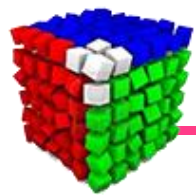


Decomposição da Luz

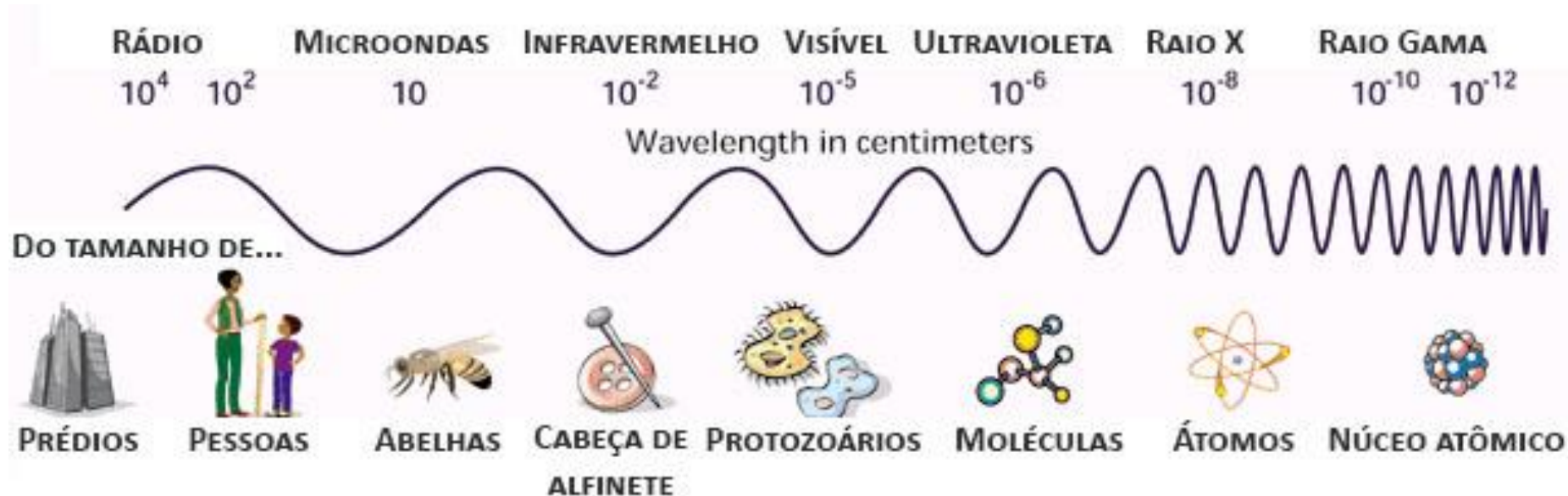
- Decomposição da luz através de um prisma

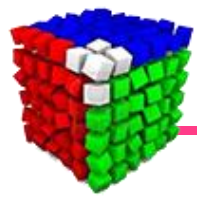
Radio	Far IR, Micro-wave	IR	UV	x-ray γ -ray
-------	-----------------------	----	----	------------------------



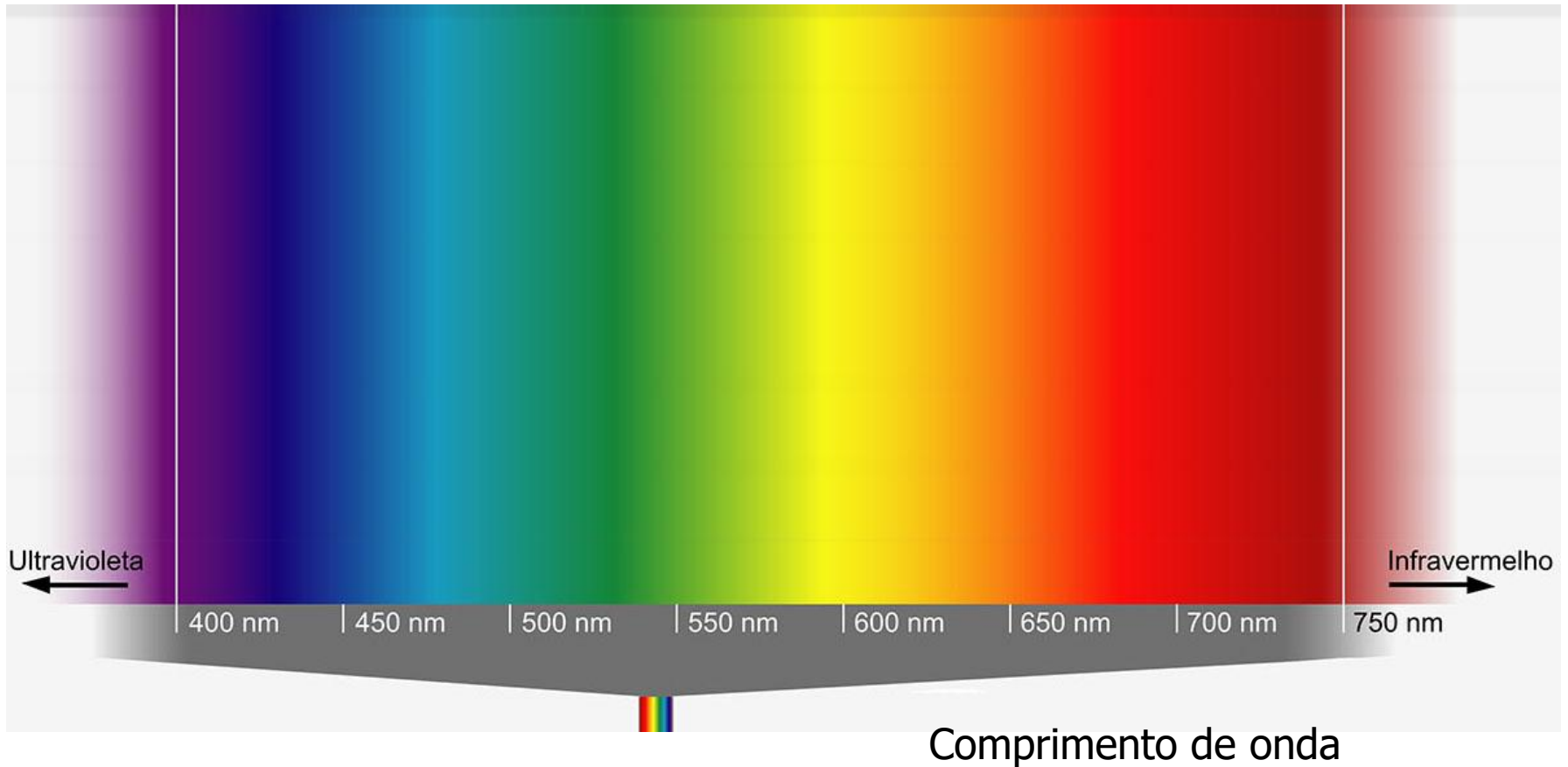


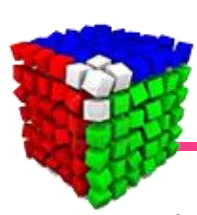
Espectro Eletromagnético





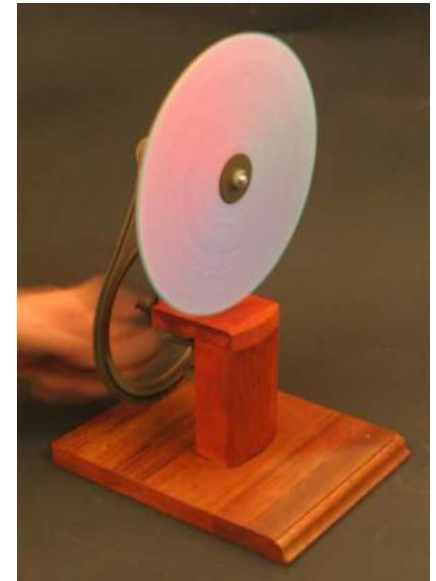
Cores do Espectro



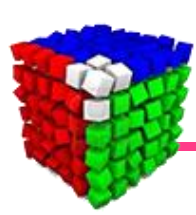


Experiências de Newton

- Newton usou 7 cores (vermelho, laranja, amarelo, verde, azul, índigo e violeta) para dividir o espectro, em analogia às 7 notas da escala musical
- Disco de Newton



<https://www.youtube.com/watch?v=b3NXsgjPSQo>

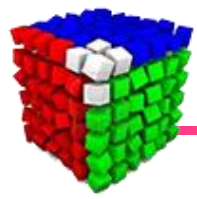


Cores - Aditivas e Subtrativas

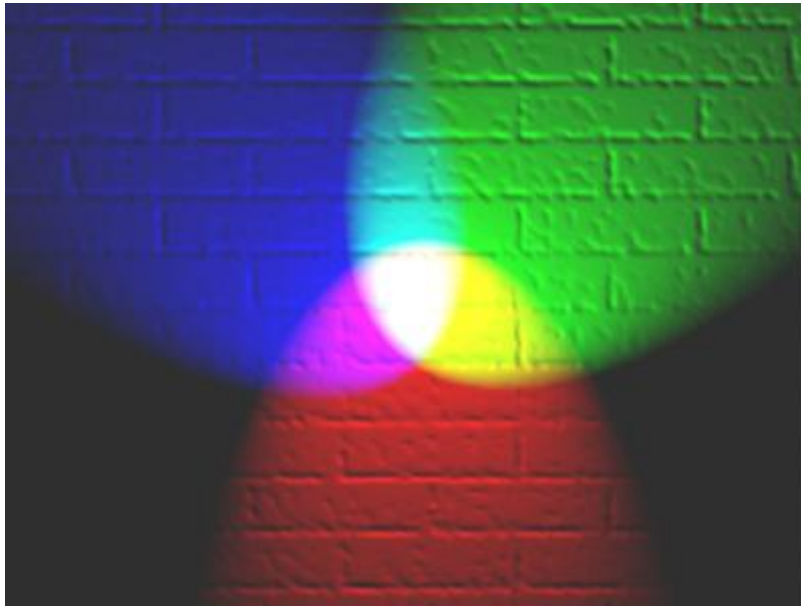
- O processo de formação **aditivo** é caracterizado pela soma de duas ou mais ondas eletromagnéticas do espectro visível, de comprimentos de onda diferentes, formando uma nova cor
- O processo de formação **subtrativo**, por sua vez, ocorre quando a luz passa por um filtro que absorve determinados comprimentos de onda e permite a passagem de outros

Cores Aditivas e Subtrativas





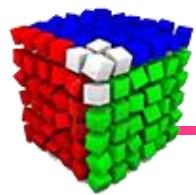
Cores - Aditivas e Subtrativas



Aditivas (luz)



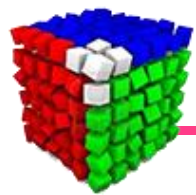
Subtrativas
(pigmentos ou filtros)



O Modelo CMY

- Modelo subtrativo
- Baseado em pigmentos
- É dependente dos dispositivos de impressão
- C - ciano
- M - magenta
- Y - amarelo

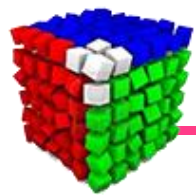




Modelo CMY ou CMYK

- O uso do pigmento preto reduz a necessidade dos outros pigmentos (ciano, magenta e amarelo)





Conversões entre RGB e CMY(K)

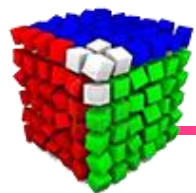
- Espaço CMY:

$$\begin{pmatrix} C \\ M \\ Y \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} R \\ G \\ B \end{pmatrix}$$

$$R = R/255$$

$$G = G/255$$

$$B = B/255$$



Conversões entre RGB e CMY(K)

- Espaço CMYK Ideal (Ford e Roberts, 1998):

$$\begin{pmatrix} C \\ M \\ Y \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} R \\ G \\ B \end{pmatrix}$$

$$R = R/255$$

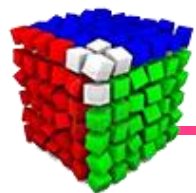
$$G = G/255$$

$$B = B/255$$

$$K = \min(C, M, Y)$$

$$\begin{pmatrix} C' \\ M' \\ Y' \end{pmatrix} = \begin{pmatrix} C \\ M \\ Y \end{pmatrix} - \begin{pmatrix} K \\ K \\ K \end{pmatrix} \bigg/ (1 - K) \quad k \neq 1$$

Observe que, nesse caso, um componente sempre terá valor nulo (o que for o mínimo para K).



Conversões entre RGB e CMY(K)

○ Espaço CMYK:

$$R = R/255$$

$$K = 1 - \max(R, G, B)$$

$$G = G/255$$

Se $k \neq 1$:

$$C = (1 - R - K)/(1 - K)$$

$$B = B/255$$

$$M = (1 - G - K)/(1 - K)$$

$$Y = (1 - B - K)/(1 - K)$$

Se $k = 1$:

$$C = (1 - R - K)$$

$$M = (1 - G - K)$$

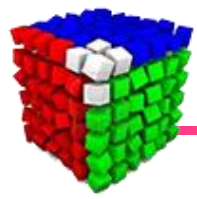
$$Y = (1 - B - K)$$

Exemplos:

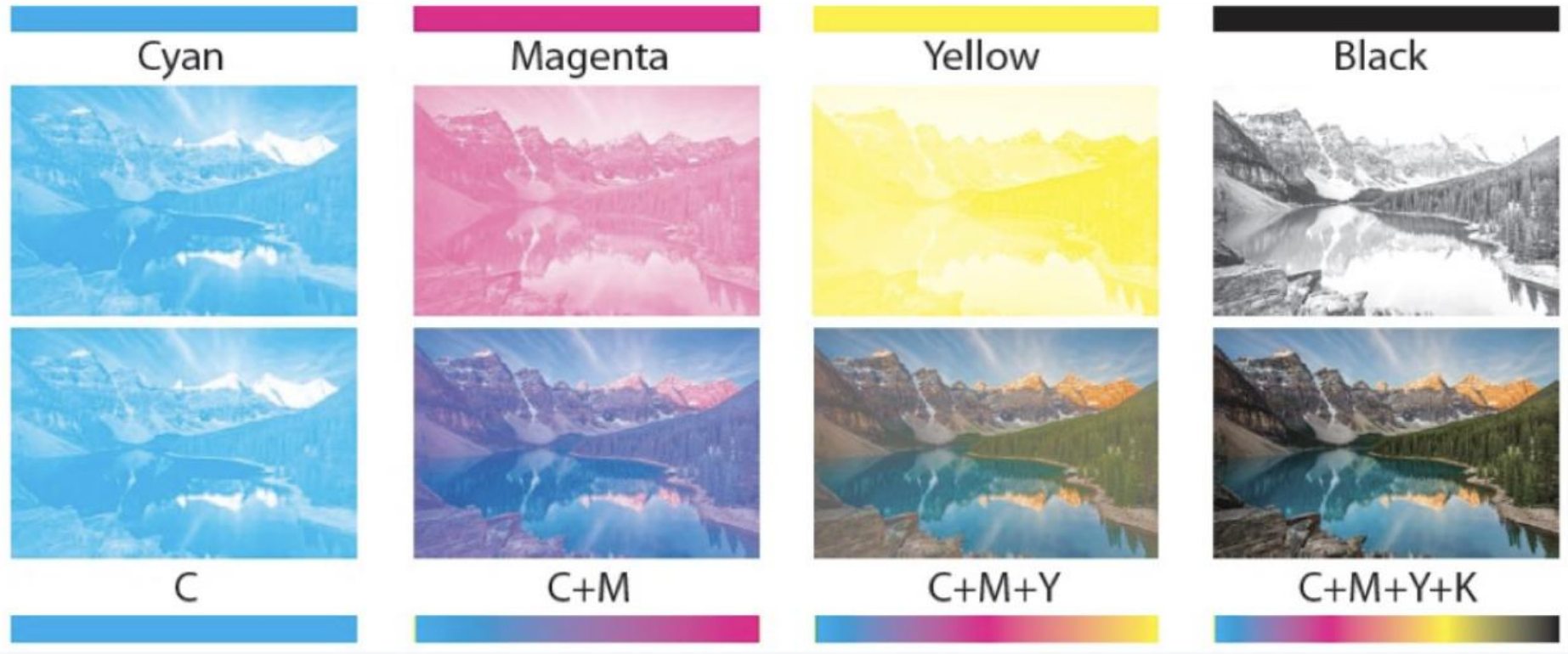
$$(120, 40, 100)_{\text{RGB}} \Rightarrow (0, 0.67, 0.17, 0.53)_{\text{CMYK}}$$

$$(255, 255, 255)_{\text{RGB}} \Rightarrow (0, 0, 0, 0)_{\text{CMYK}}$$

$$(0, 0, 0)_{\text{RGB}} \Rightarrow (0, 0, 0, 1)_{\text{CMYK}}$$

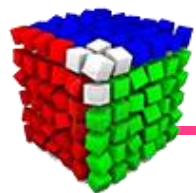


Conversões entre RGB e CMY(K)



Sistemas CIE



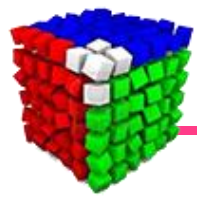


Valores Tricromáticos

- Qualquer cor C poderia ser reproduzida através das cores primárias:

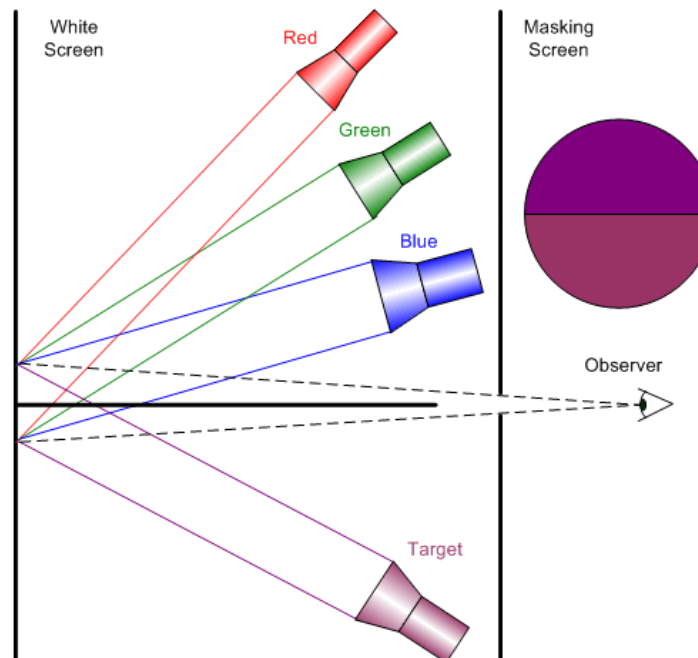
$$C = r \cdot R + g \cdot G + b \cdot B$$

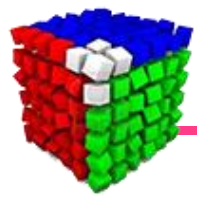
- Onde R , G e B são as três cores primárias e r , g e b são os coeficientes correspondentes



Sistema Padrão CIE-RGB

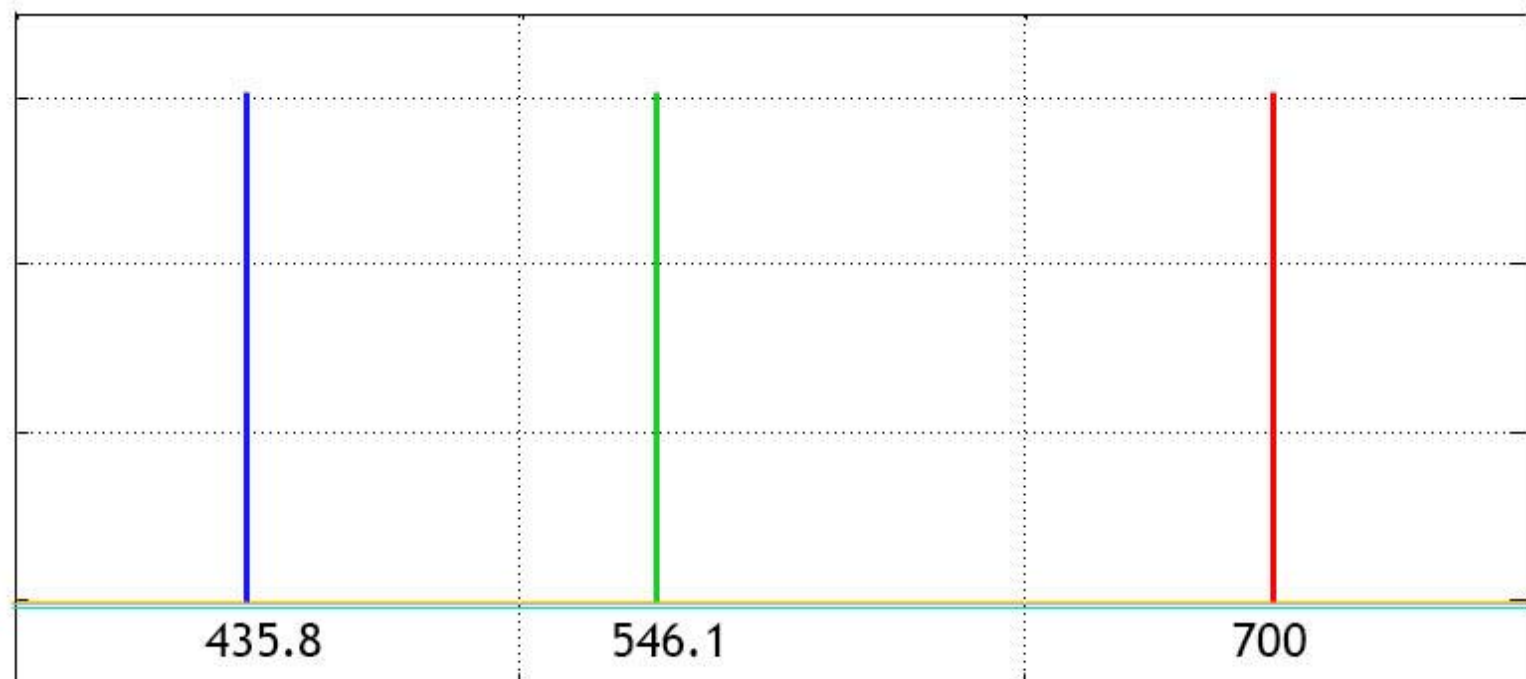
- CIE: *Commission Internationale d'Eclairage*
- Baseado em resultados experimentais
- W. David Wright e John Guild independentemente



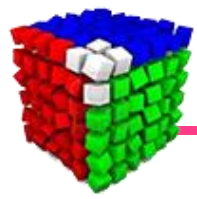


Valores Tricromáticos

Observadores padrão CIE 1931

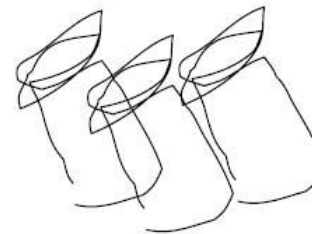
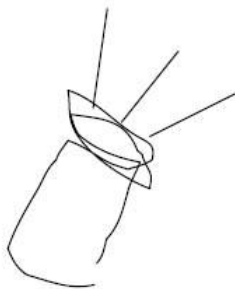


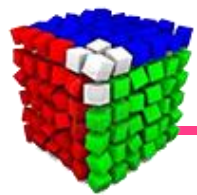
COMPRIMENTO DE ONDA (nm)



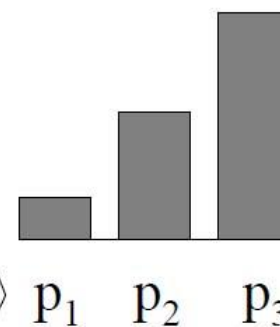
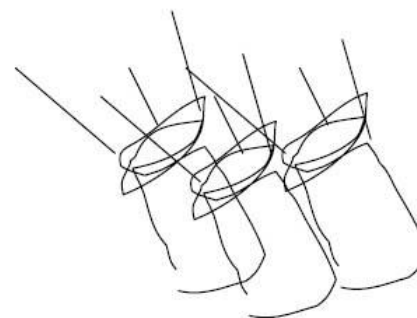
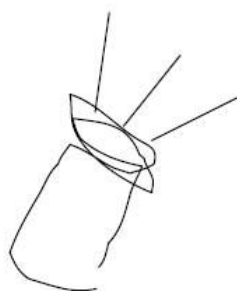
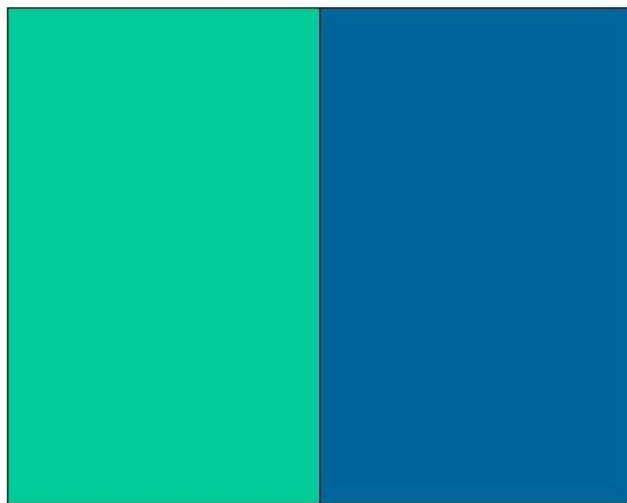
Experimentos CIE-RGB

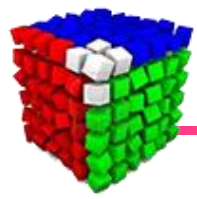
- Experimento realizado para “cada cor”



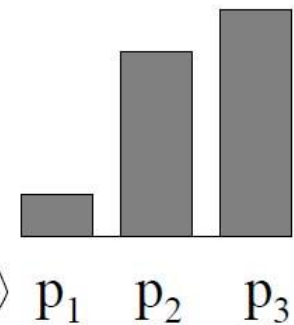
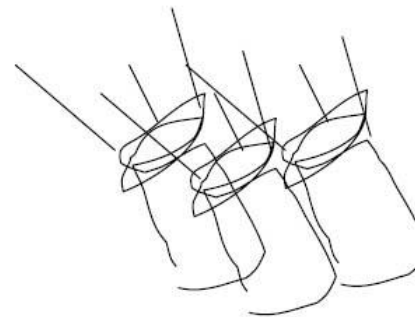
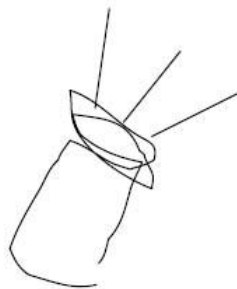
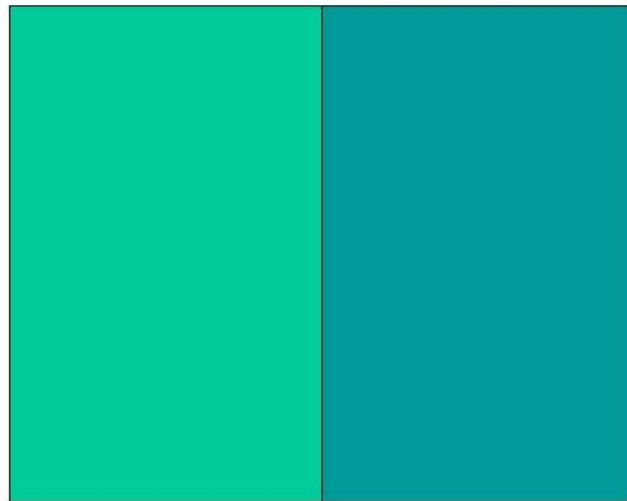


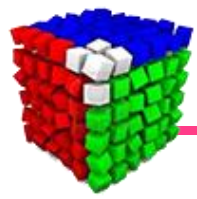
Experimentos CIE-RGB



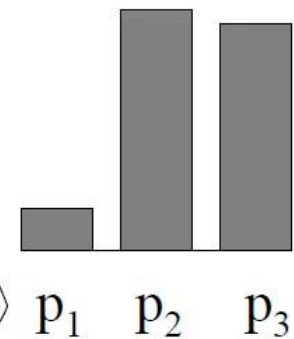
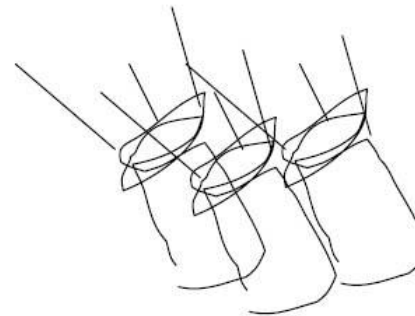
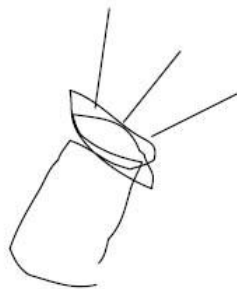
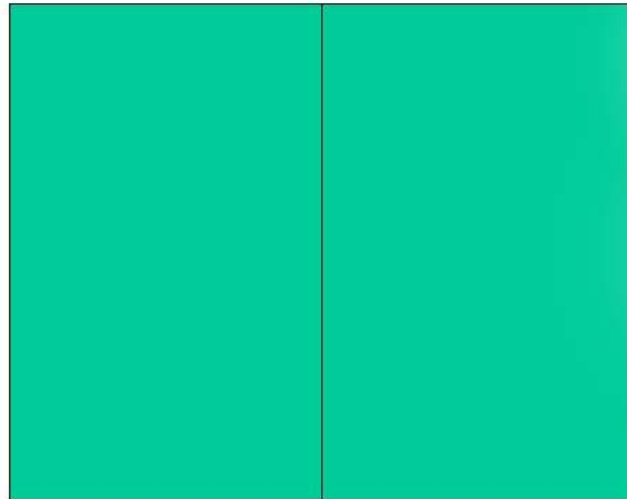


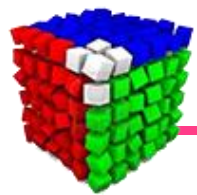
Experimentos CIE-RGB



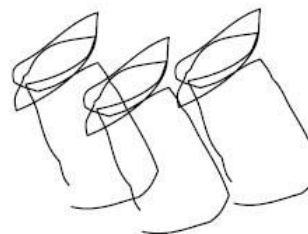
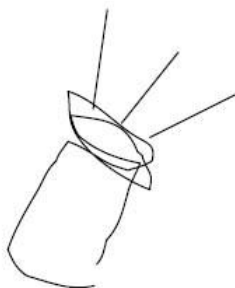
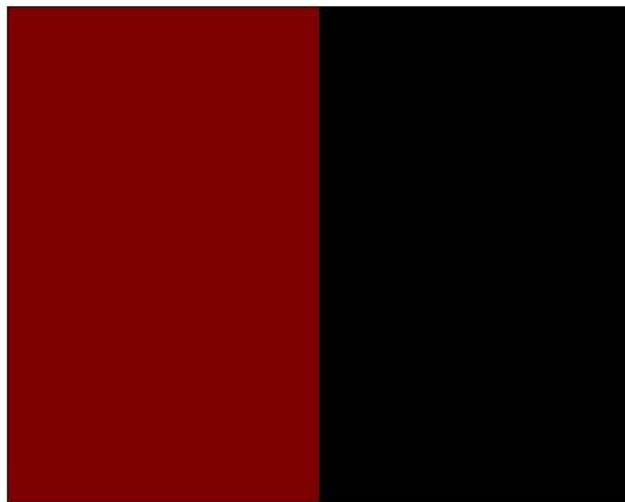


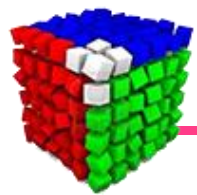
Experimentos CIE-RGB



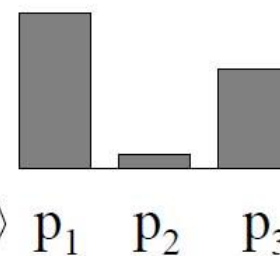
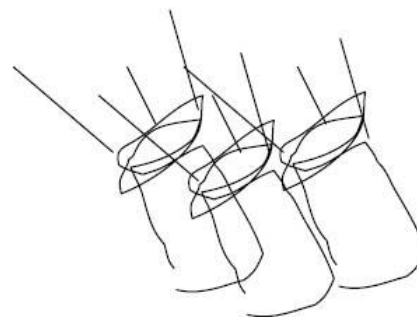
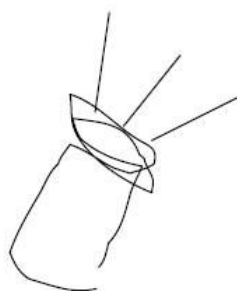
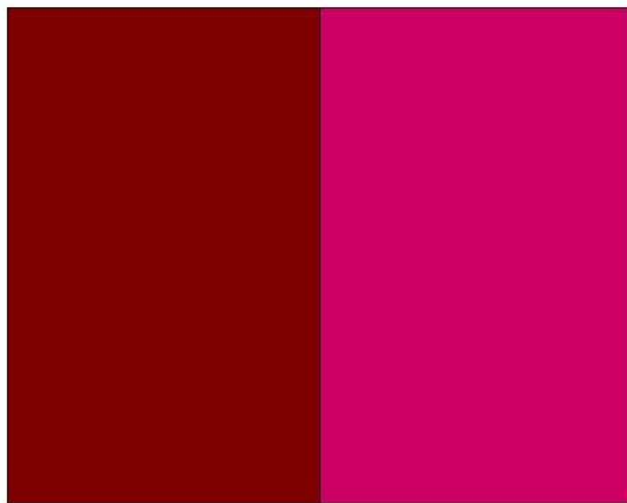


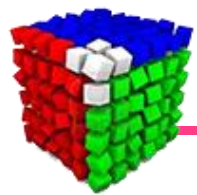
Experimentos CIE-RGB





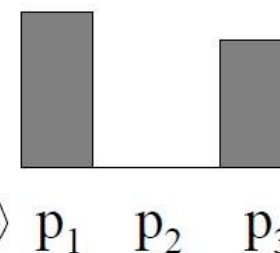
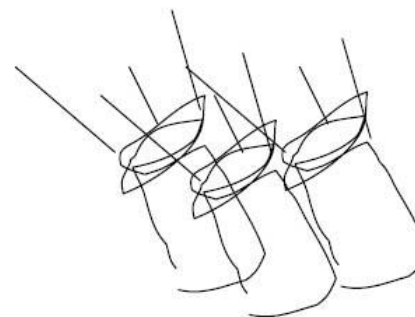
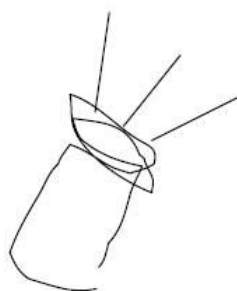
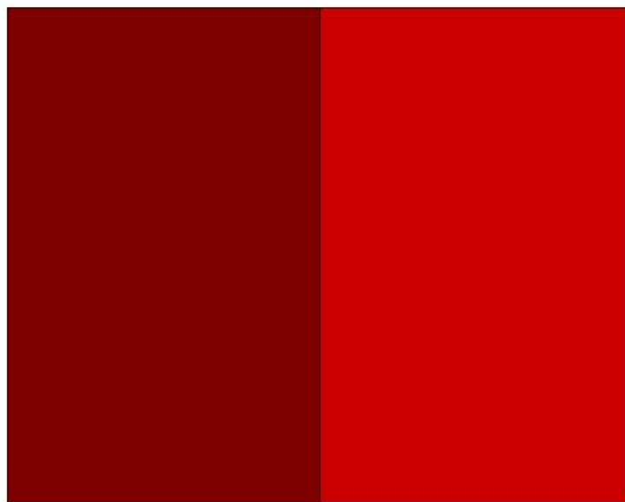
Experimentos CIE-RGB

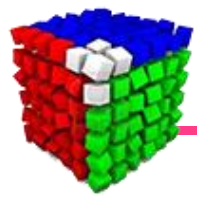




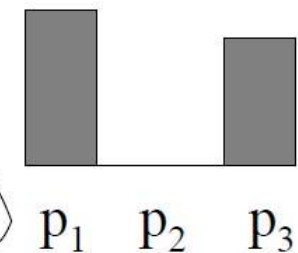
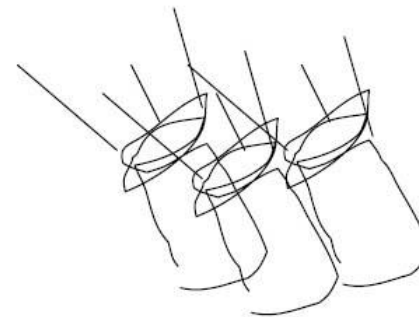
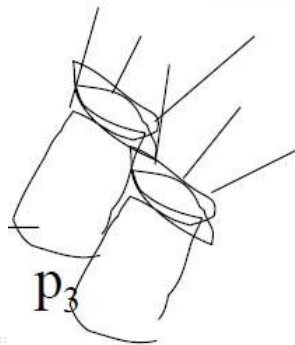
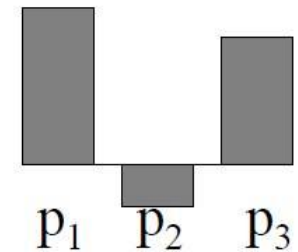
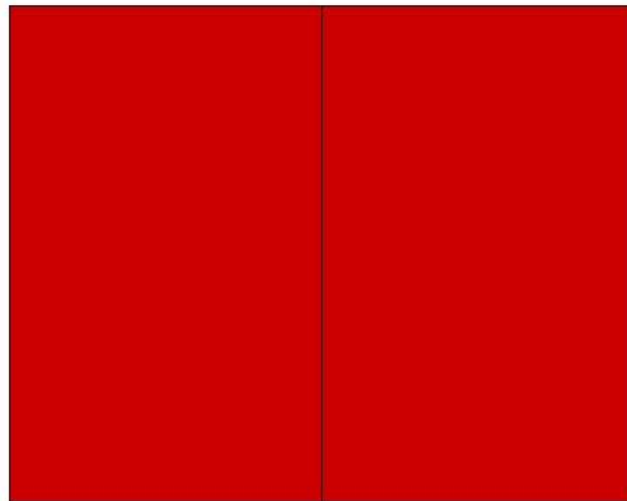
Experimentos CIE-RGB

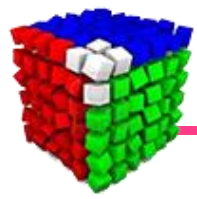
Problema....





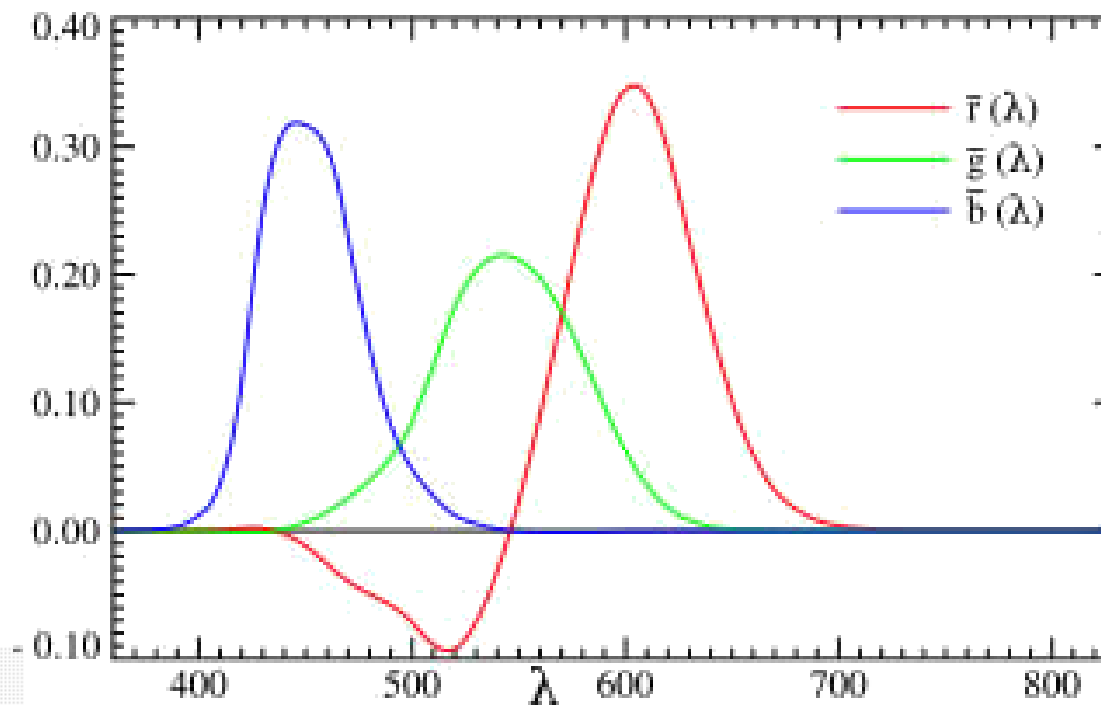
Experimentos CIE-RGB

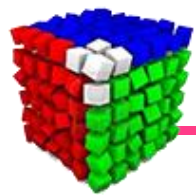




Sistema Padrão CIE-RGB

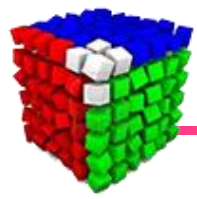
- As curvas representam as quantidades de primárias necessárias para reproduzir a cor de referência





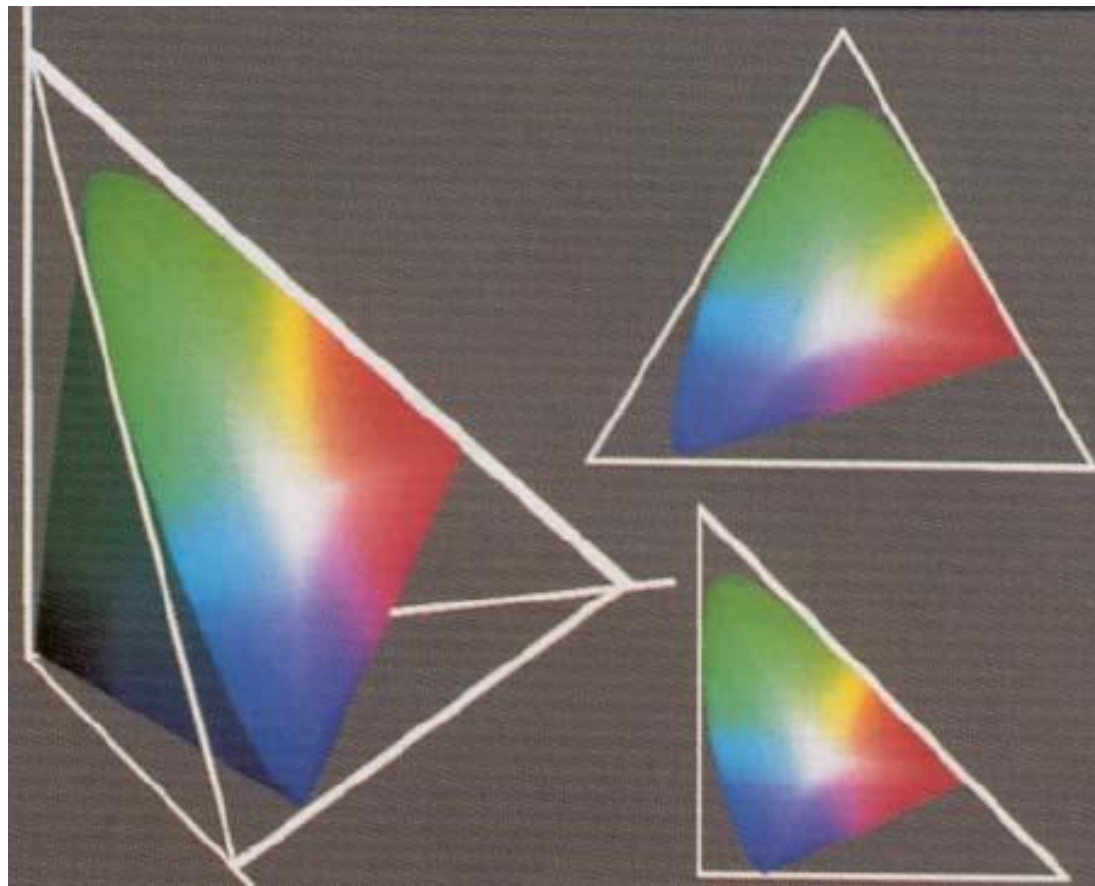
Sistema Padrão CIE-XYZ

- Algumas cores não puderam ser reproduzidas usando apenas valores positivos no CIE-RGB
- Em 1931, a CIE definiu três novas primárias: X, Y e Z
- Somente valores positivos
- Y (luminância), X, Z (croma)



Sistema Padrão CIE- XYZ

- Vistas do plano $X+Y+Z=1$



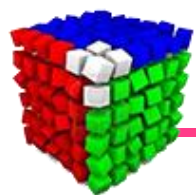
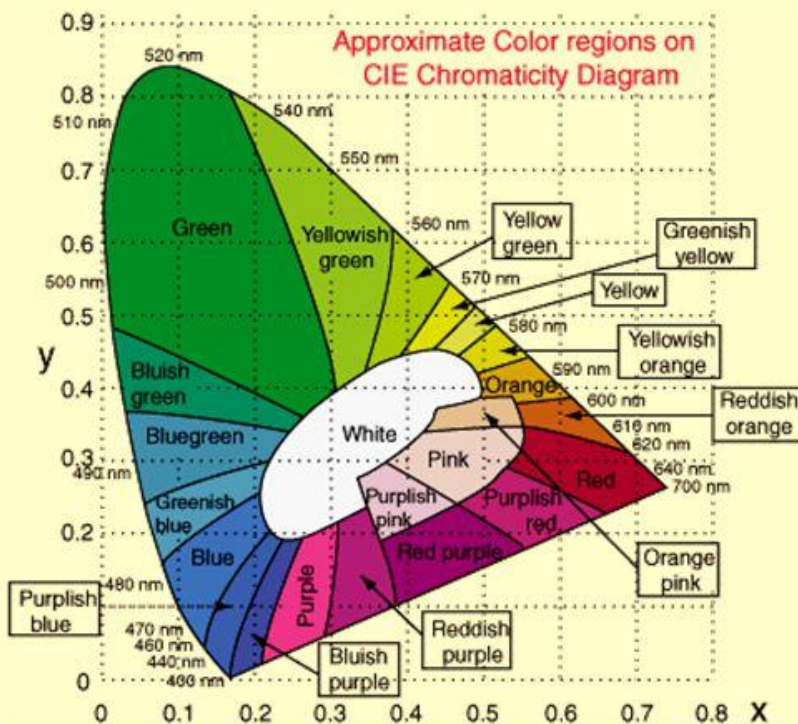


Diagrama de Cromaticidade

Interpretação do diagrama de cromaticidade CIE



Color name	Red	Green	Blue
Red	191	27	75
Pink	245	220	208
Reddish orange	216	119	51
Orange pink	240	204	162
Orange	228	184	29
Yellowish orange	231	224	0
Yellow	234	231	94
Greenish yellow	235	233	0
Yellow green	185	214	4
Yellowish green	170	209	60
Green	0	163	71
Bluish green	24	162	121
Bluegreen	95	164	190
Greenish blue	110	175	199
Blue	92	138	202
Purplish blue	88	121	191
Bluish purple	92	102	177
Purple	246	85	158
Reddish purple	196	64	143
Purplish pink	243	208	219
Red purple	175	35	132
Purplish red	209	65	136
White	255	255	255

Diagrama que caracteriza as cores por um parâmetro Y de Luminância e duas coordenadas de cores x e y que especifica o ponto no diagrama de cromaticidade.

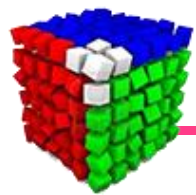
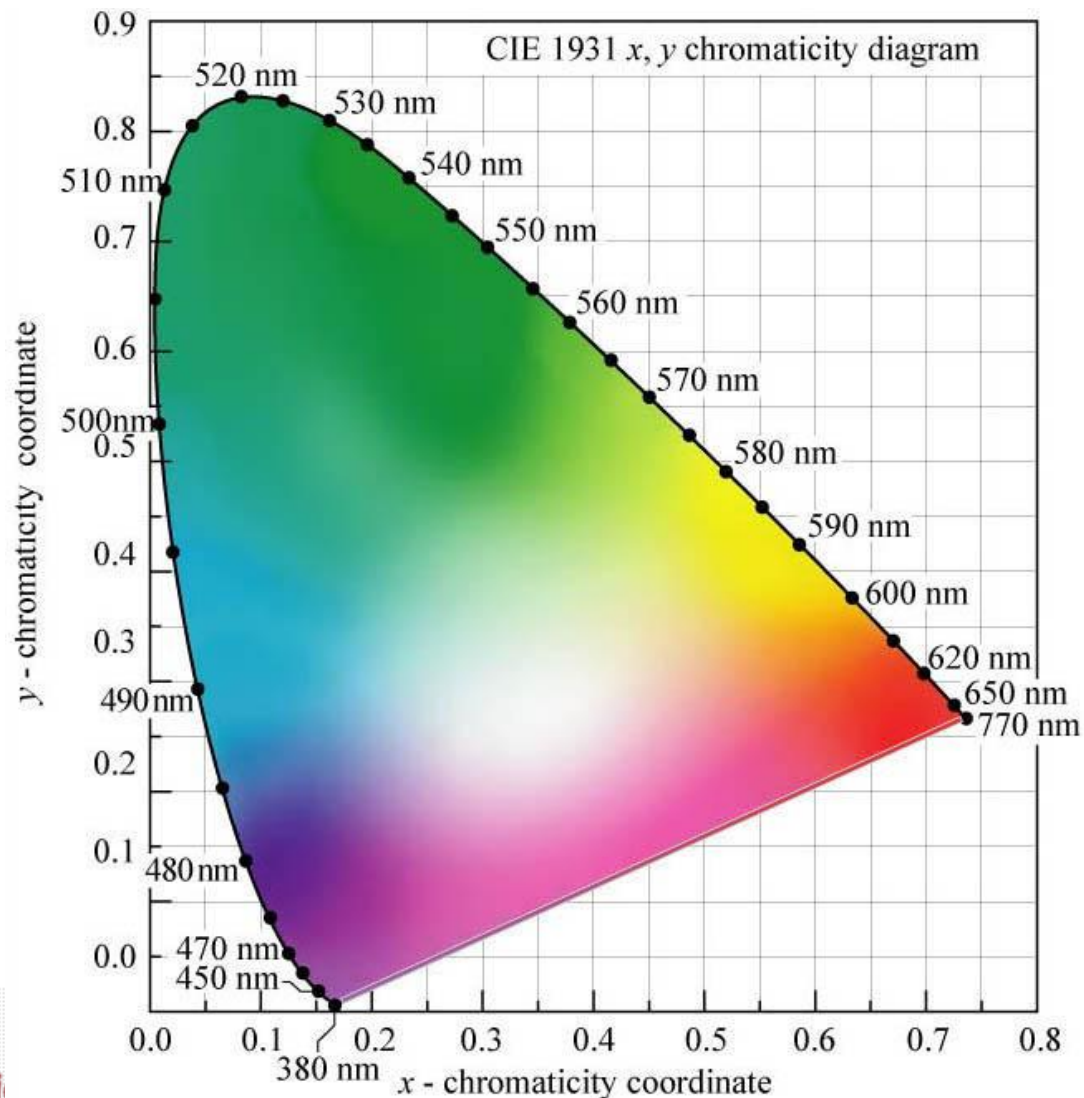


Diagrama de Cromaticidade



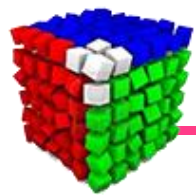
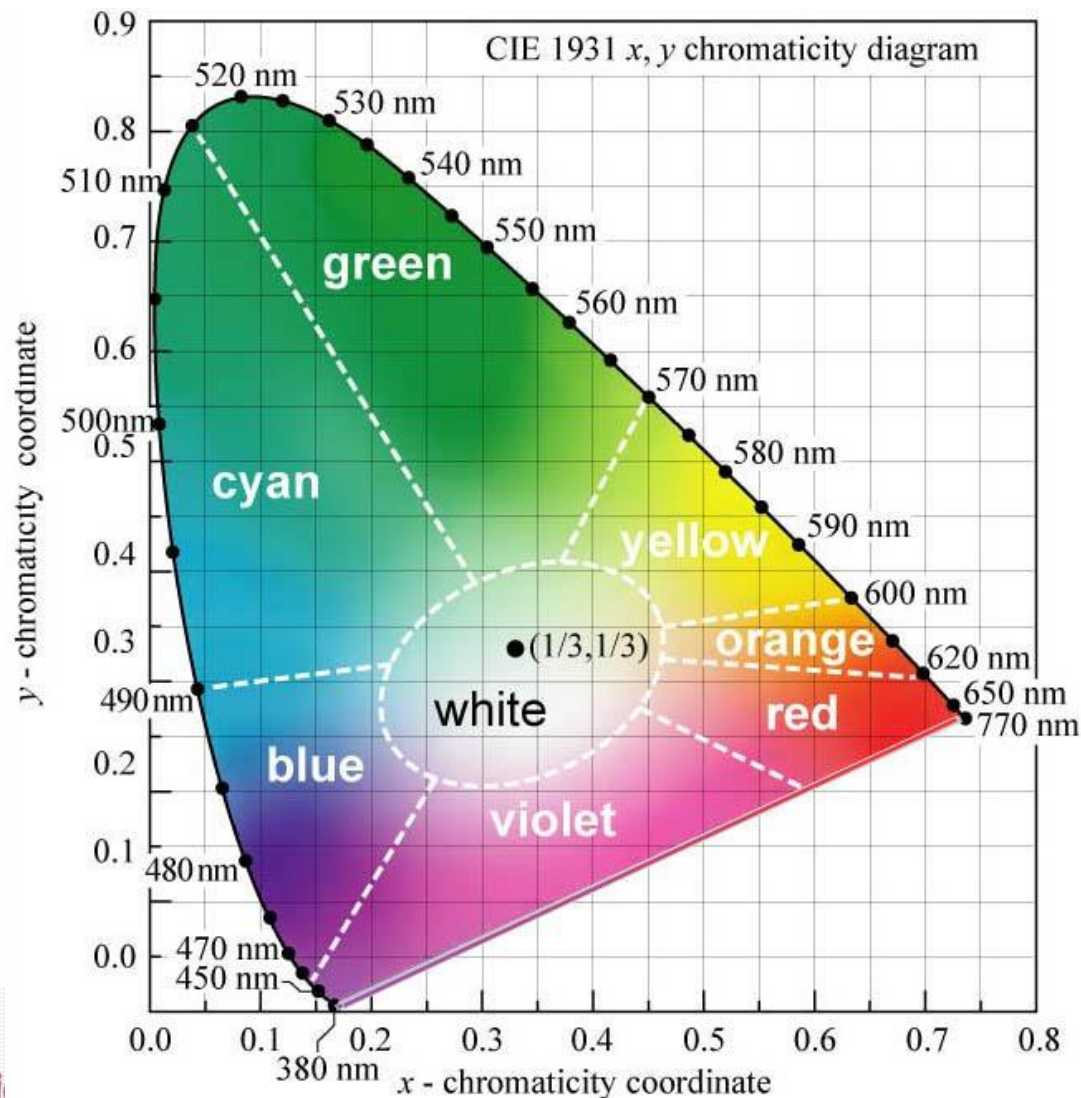


Diagrama de Cromaticidade



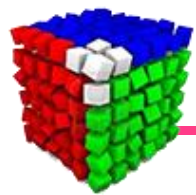
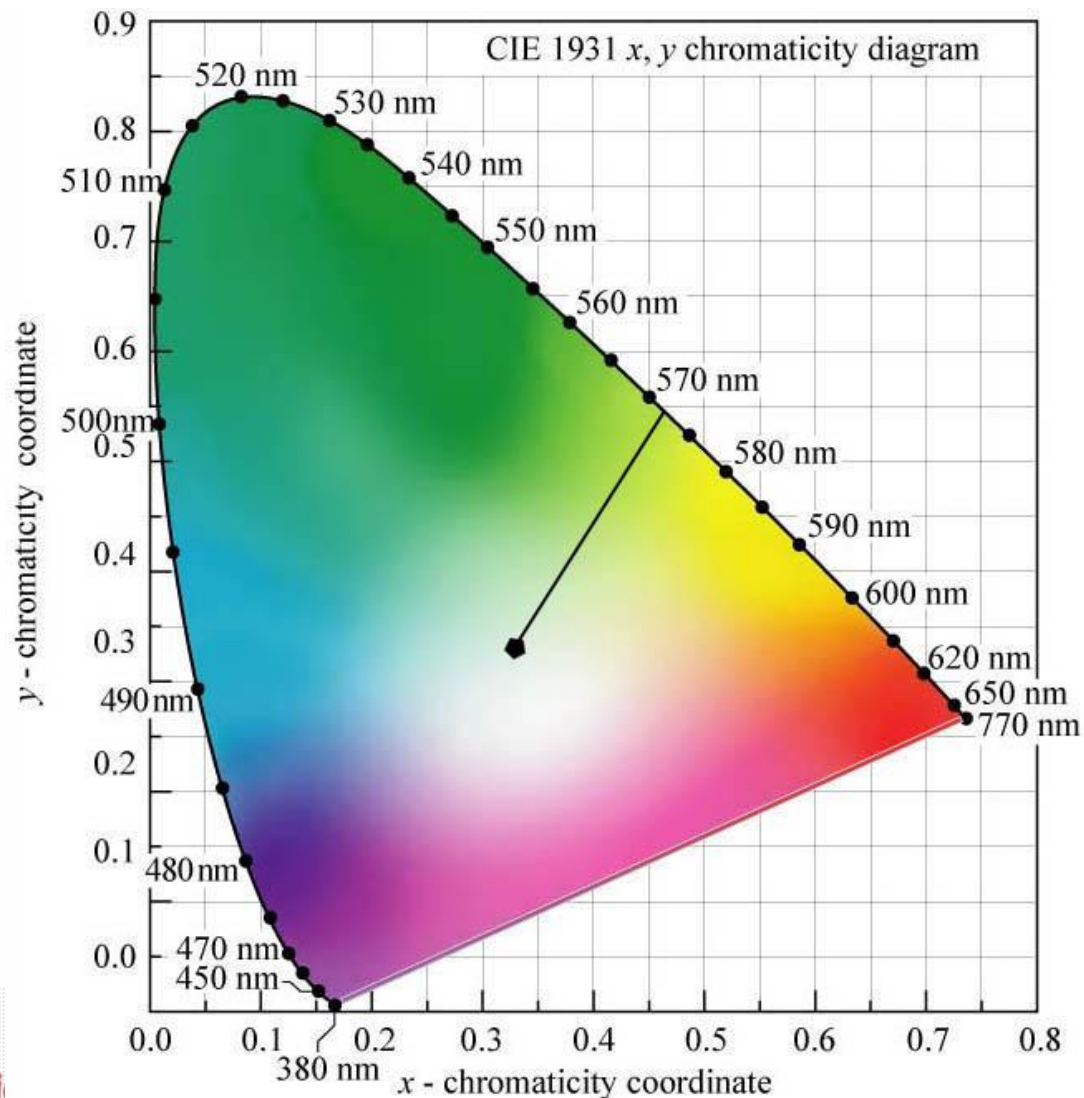


Diagrama de Cromaticidade



Saturação

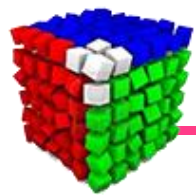
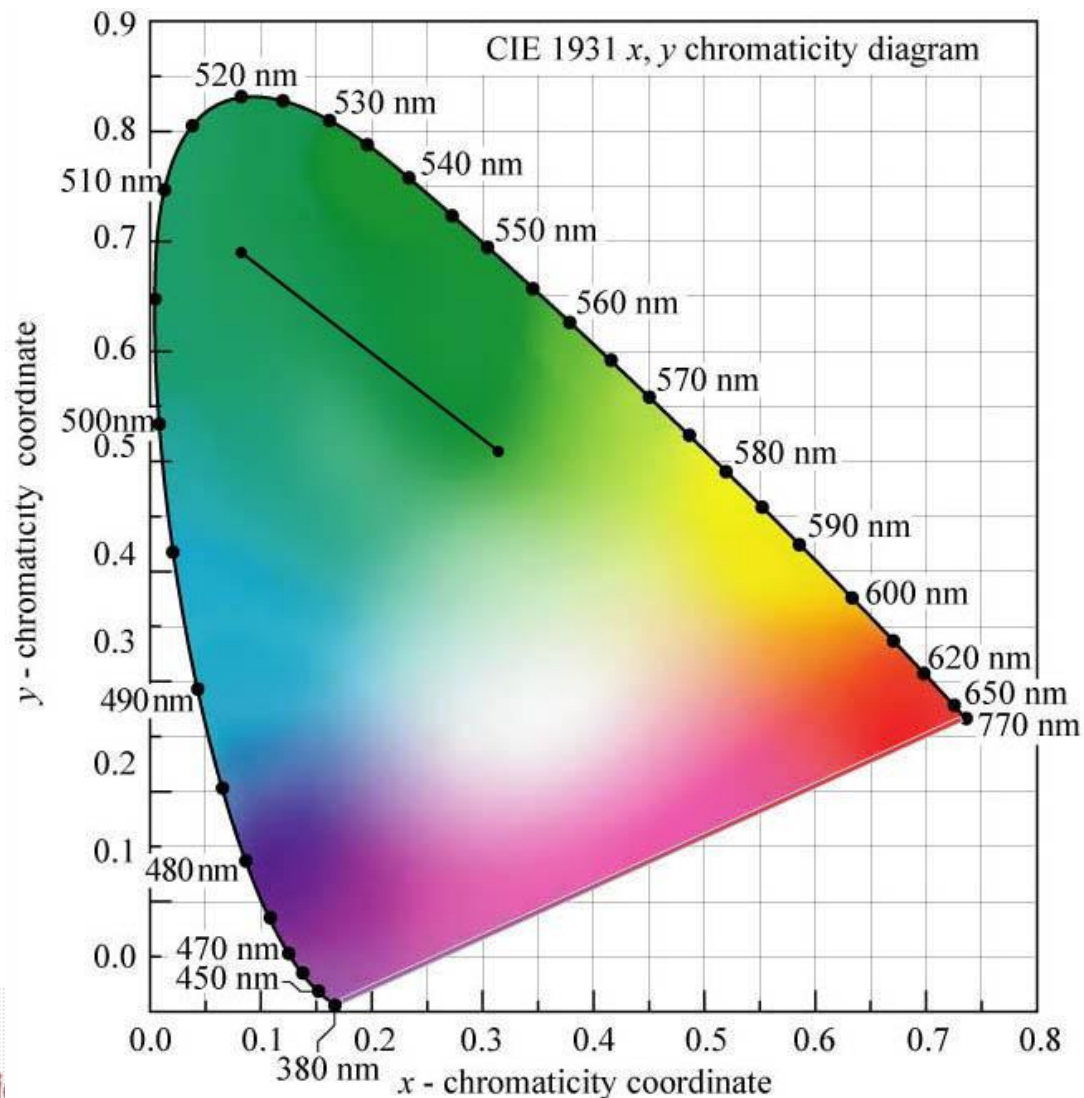


Diagrama de Cromaticidade



Combinação
De cores

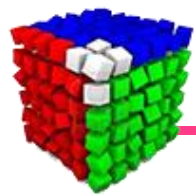
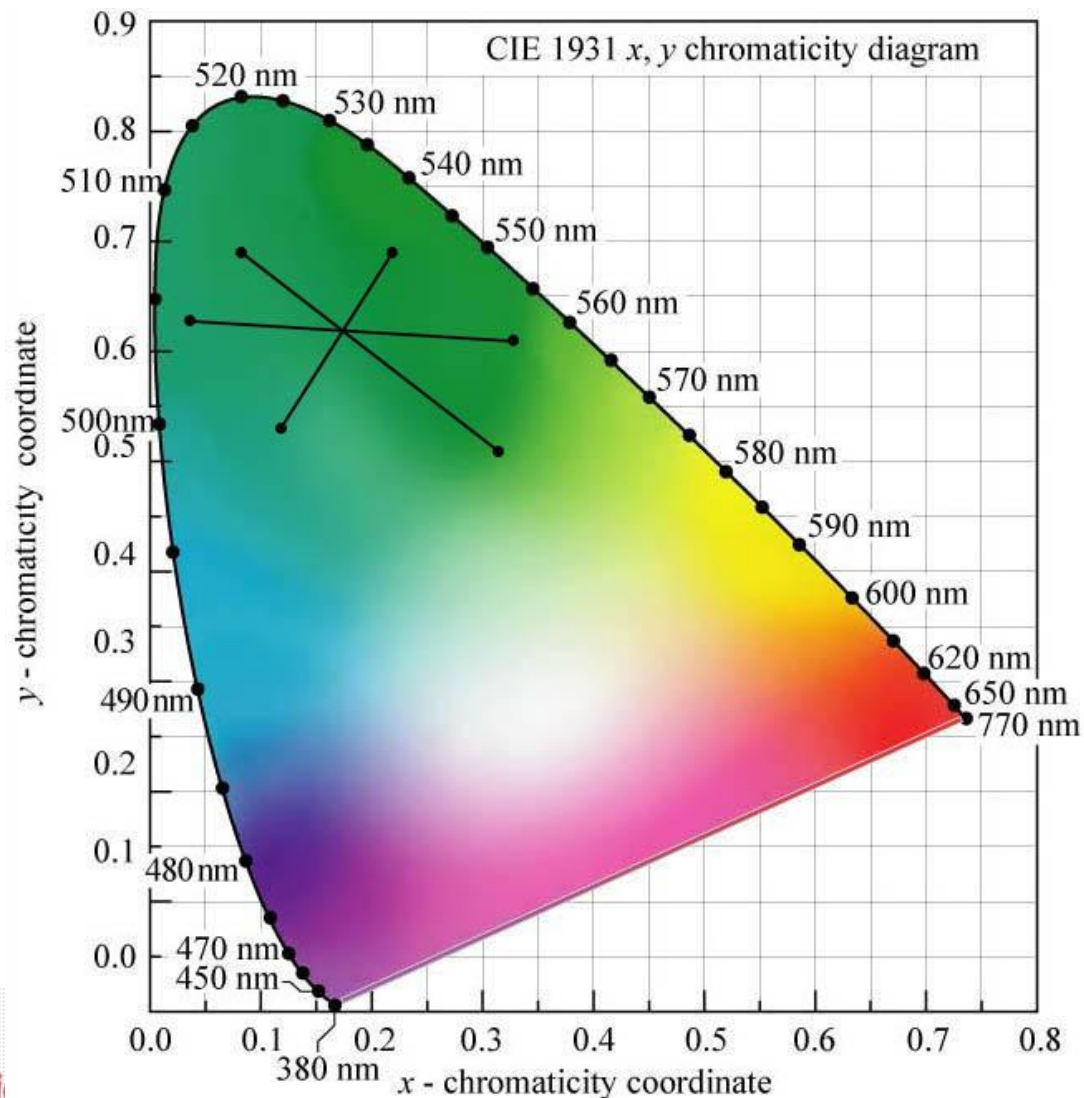


Diagrama de Cromaticidade



Metâmeros:
Cores que
parecem
iguais

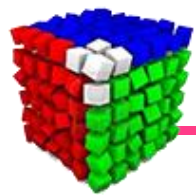
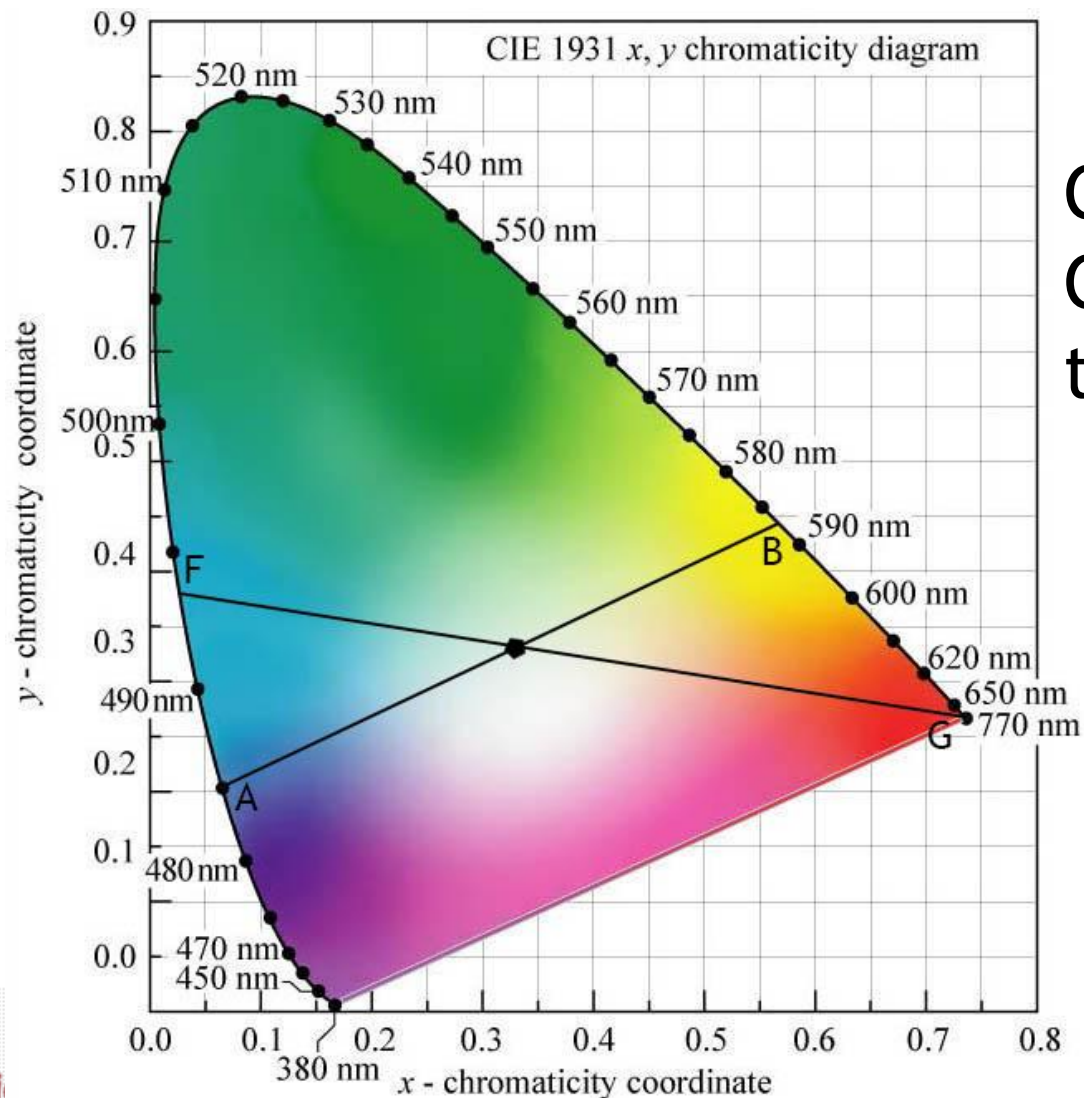


Diagrama de Cromaticidade



Cores
Complementares

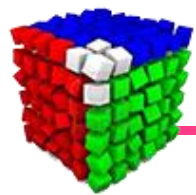
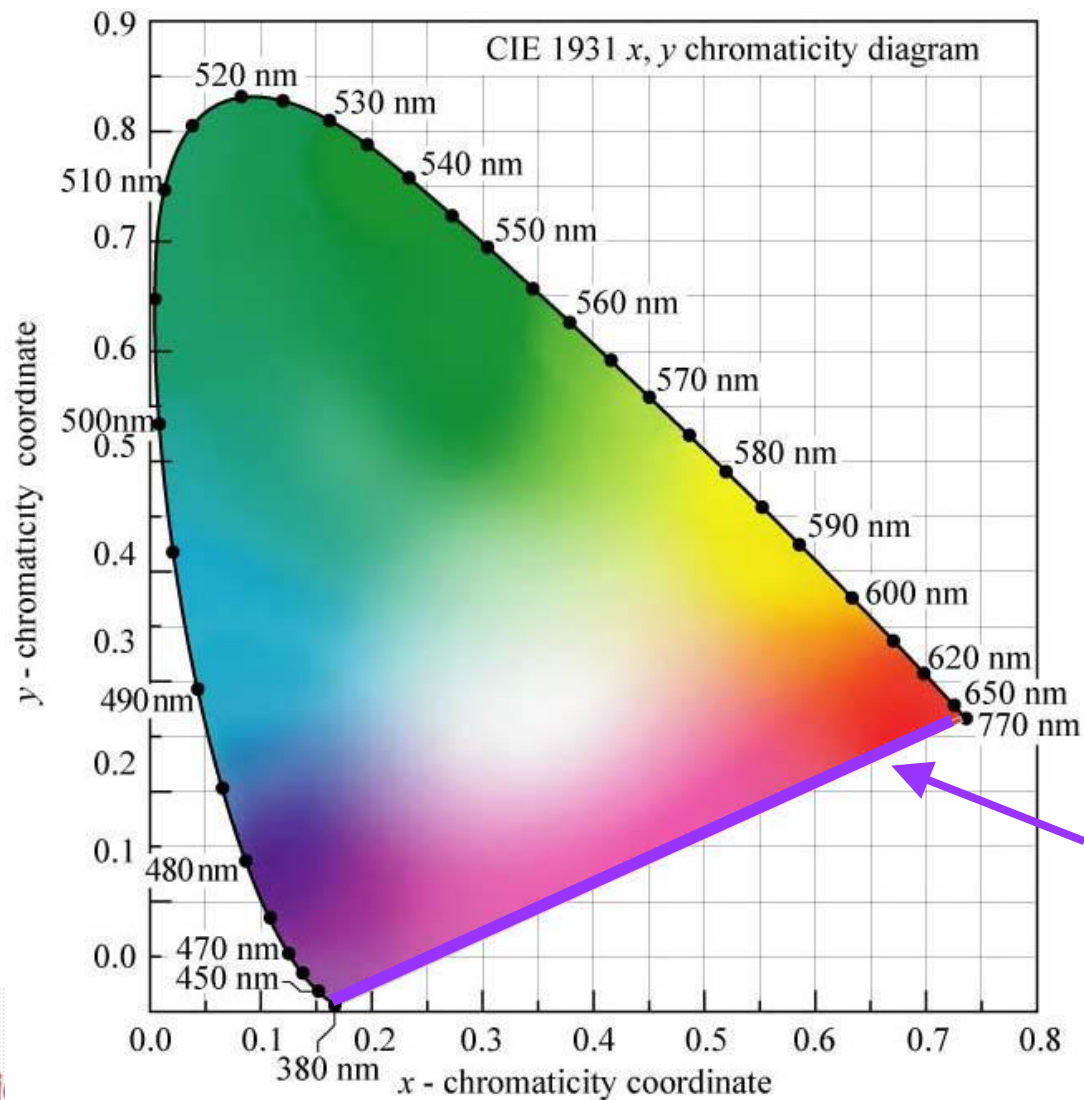
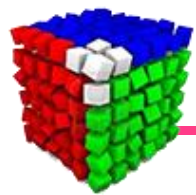


Diagrama de Cromaticidade



Linha púrpura



Conversão entre RGB e XYZ

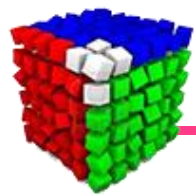
$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} 0.4124 & 0.3576 & 0.1805 \\ 0.2126 & 0.7152 & 0.0722 \\ 0.0193 & 0.1192 & 0.9505 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

$$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} 3.2405 & -1.5371 & -0.4985 \\ -0.9693 & 1.8706 & 0.0416 \\ 0.0556 & -0.2040 & 1.0572 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$

Ex.: $Y = 0.2126 \cdot R + 0.7152 \cdot G + 0.0722 \cdot B$

Sistema Computacional de Cores

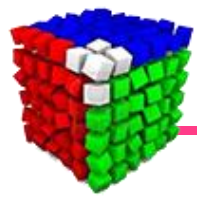




Sistema Computacional de Cores

- Cada tom (R, G e B) é armazenado em 1B
- Resolução de Cor:
 - 2 cores (1 bit)
 - 16 cores (4 bits)
 - 256 cores (8 bits = 1 Byte)
 - 16 milhões de cores (24 bits = 3 Bytes)

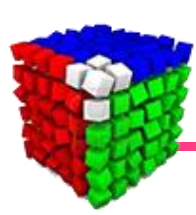
Paleta de Cores



Sistema Computacional de Cores

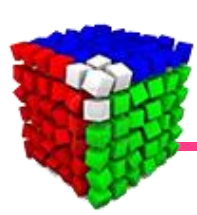
■ Sistema RGB

NUMBERS								
R	255	R	102	R	51			
G	0	G	102	G	204			
B	0	B	255	B	153			
R	255	R	255	R	51			
G	255	G	0	G	204			
B	102	B	204	B	255			
R	51	R	51	R	255			
G	51	G	51	G	153			
B	0	B	153	B	153			



Tons de cinza

- Níveis de brilho
- Níveis de cinza
- Conversão de um modelo (RGB, por exemplo) para uma representação de brilho da cor
 - Ex.: $C = \lceil 0,29R + 0,59G + 0,11B \rceil$



Tons de cinza

■ Exemplos de conversão



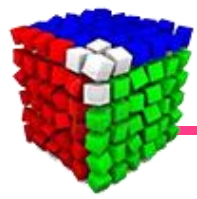
Original



Média



Pesos



Tons de cinza

■ Exemplos de conversão



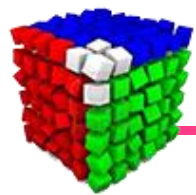
Red



Green

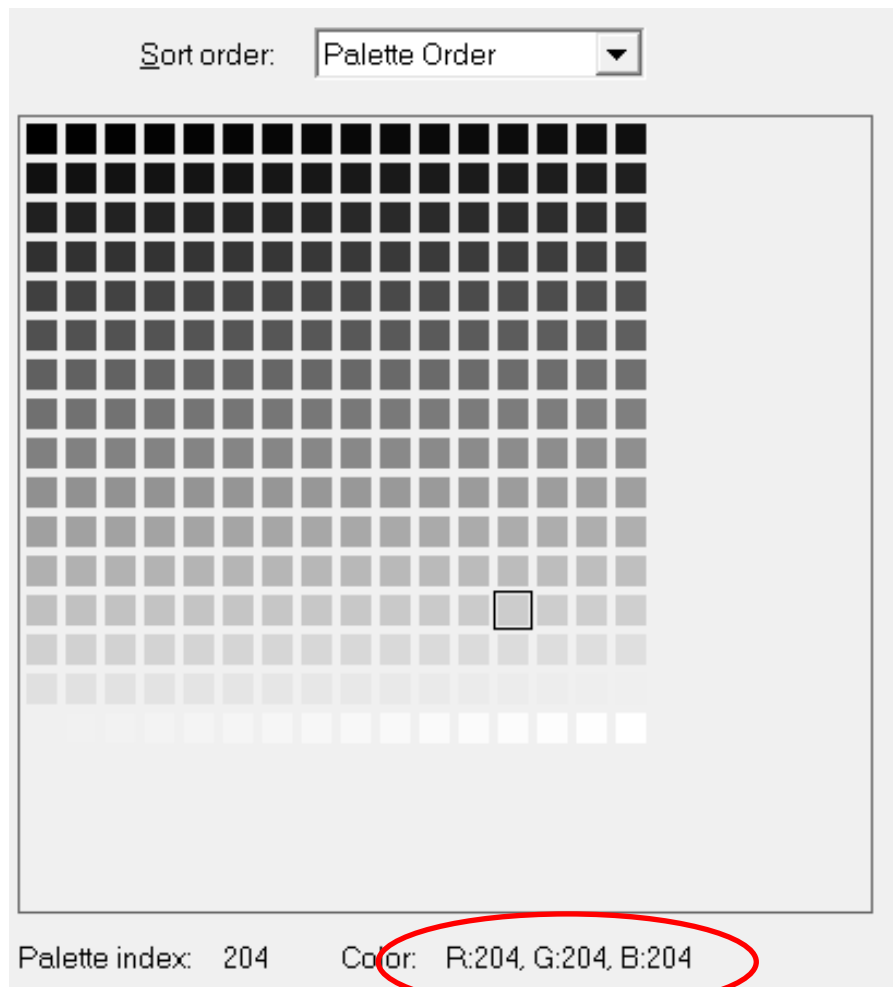


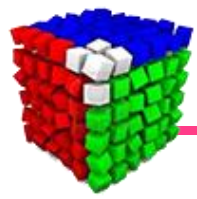
Blue



Tons de cinza

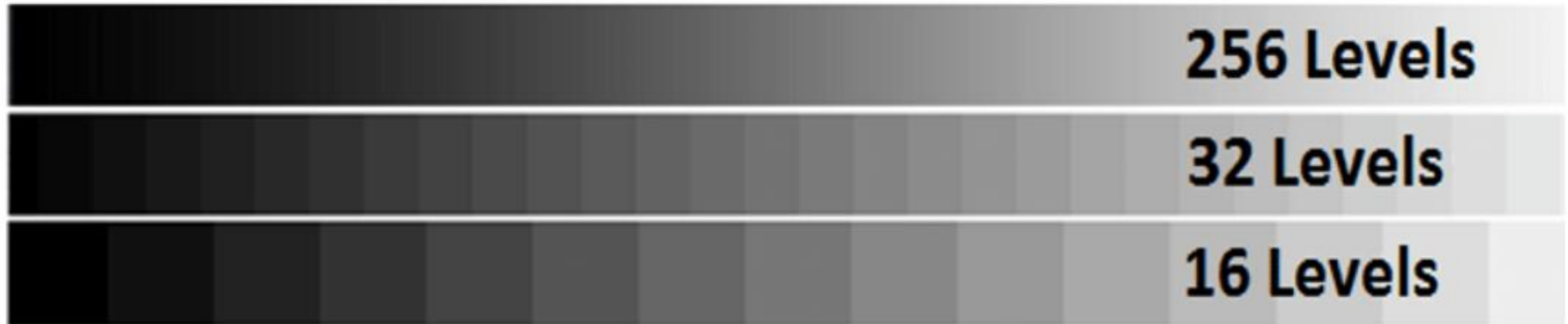
- Tons de cinza (no RGB)

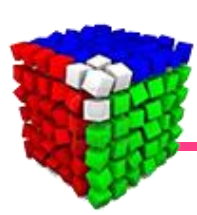




Tons de cinza

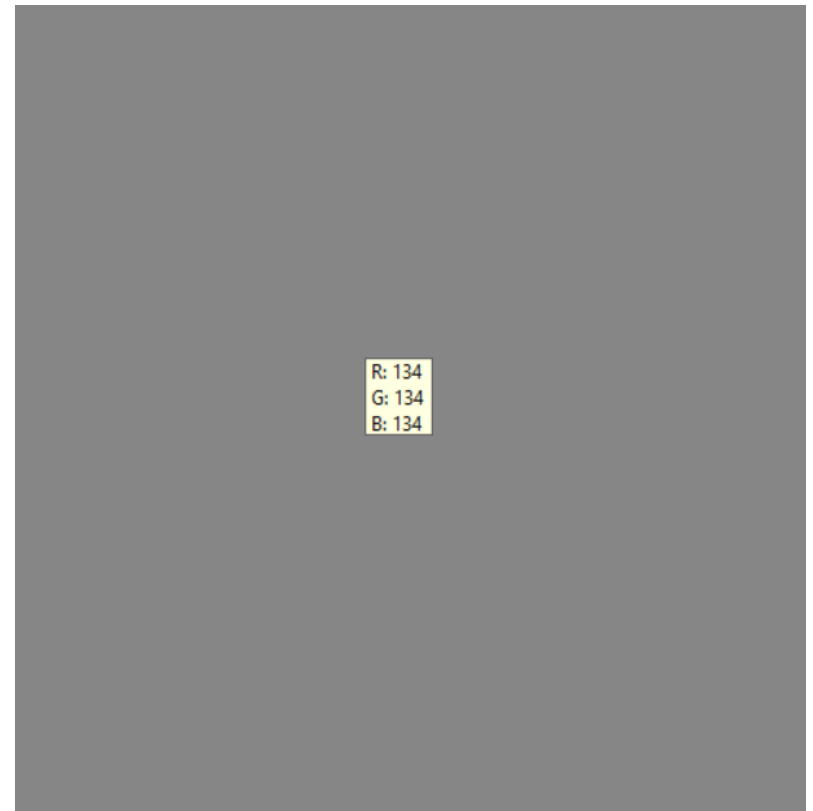
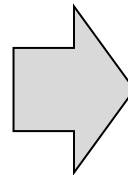
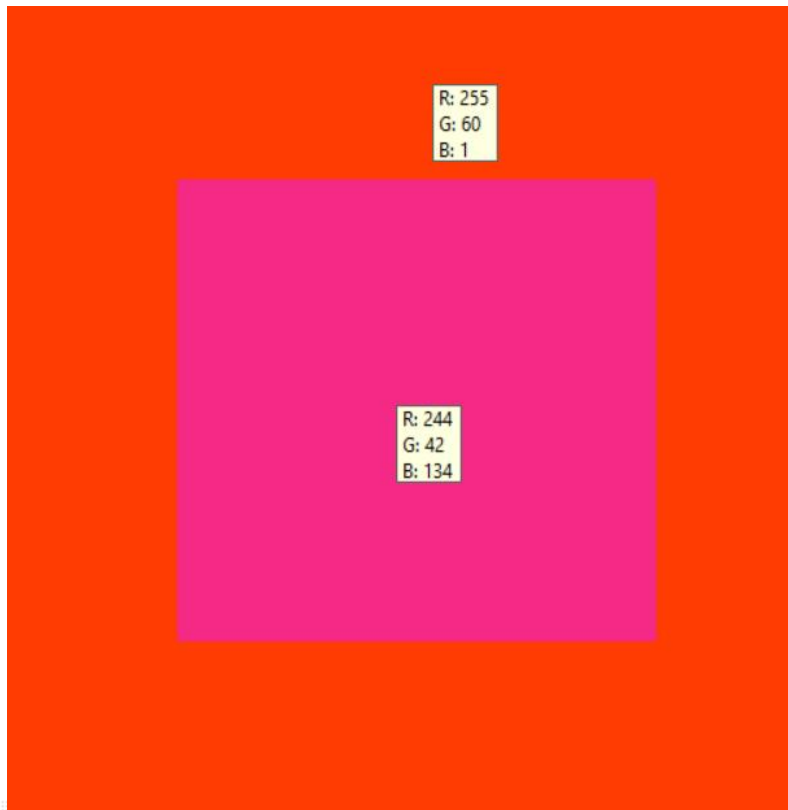
- Tons de cinza (no RGB)

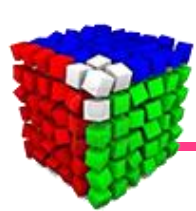




Tons de cinza

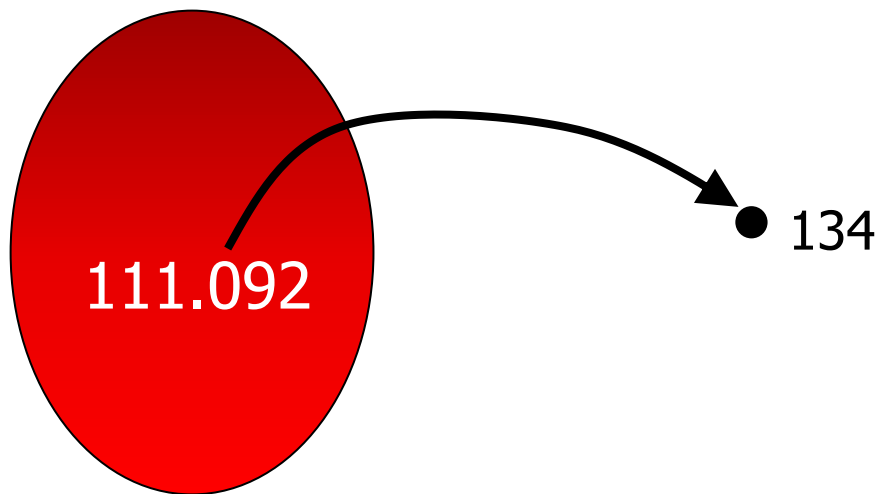
■ Cores metaméricas

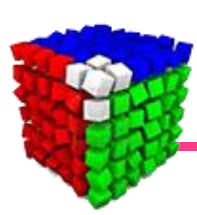




Tons de cinza

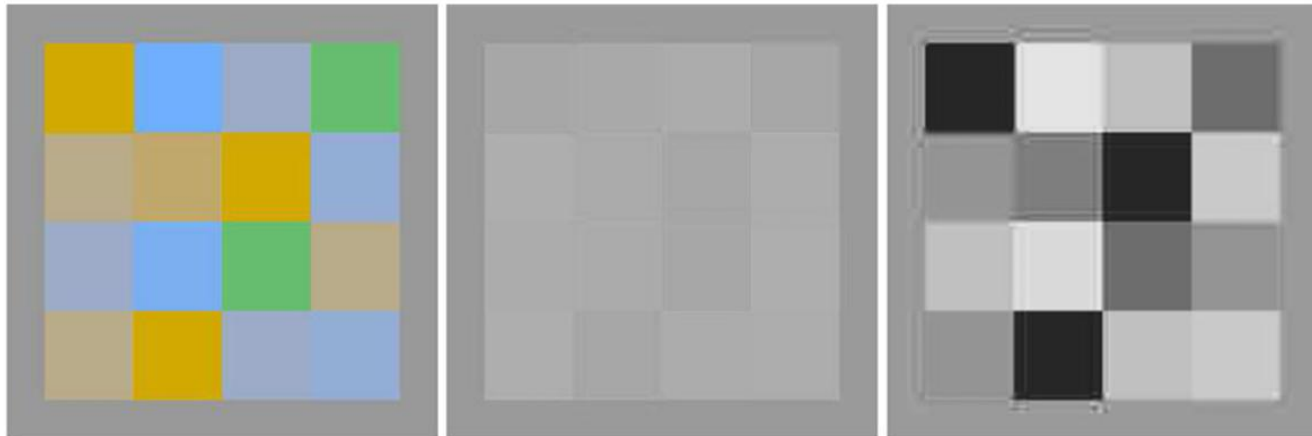
- Cores metaméricas
 - Diferentes cores são mapeadas em um tom apenas
 - Nesse caso anterior, 111.092 cores (RGB) são mapeadas no tom de cinza 134





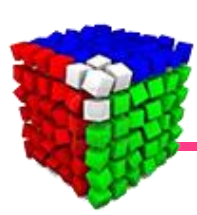
Tons de cinza

- Tema ainda em estudo e desenvolvimento
 - A new perspective on decolorization: feature-preserving decolorization, Orhan Akbulut
 - Signal, Image and Video Processing (2021)
 - <https://doi.org/10.1007/s11760-020-01802-4>



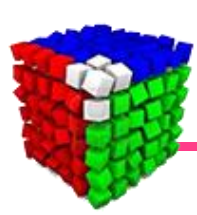
Modelos de Interface





Modelos de Interface

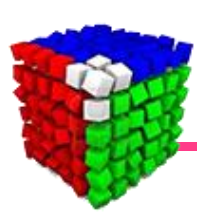
- Os modelos de interface, ou baseados na percepção, são modelos que são caracterizados, no geral, por componentes que descrevem a matiz, saturação e luminância da cor
- Difícilmente, uma pessoa entende uma cor em um modelo como o RGB



Modelo de Munsell

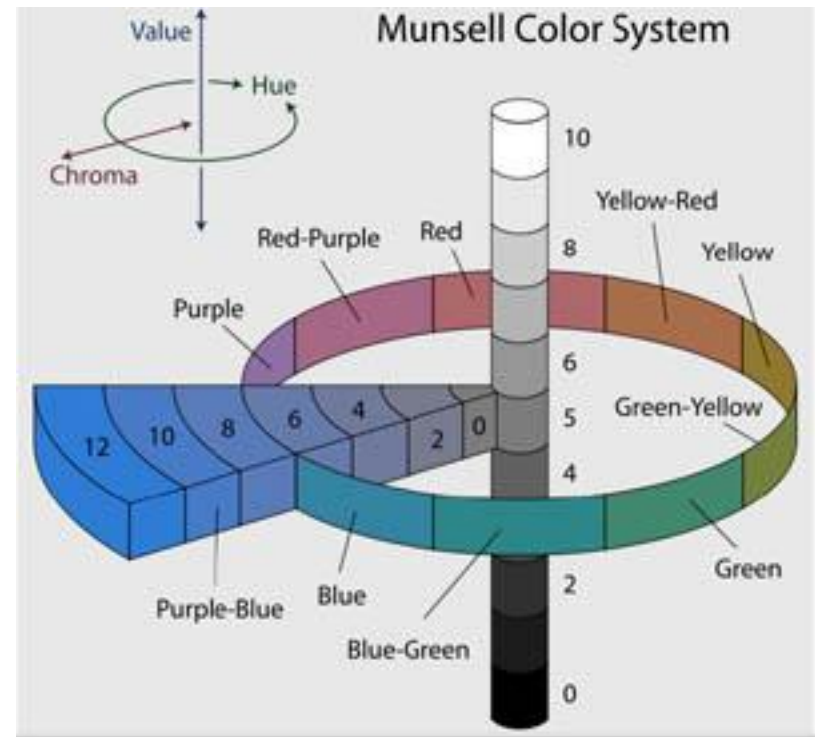
- Tentativa inicial em criar um sistema preciso para descrever cores numericamente
- O Matiz (Hue) é representado em uma ordem natural de cores na forma circular:

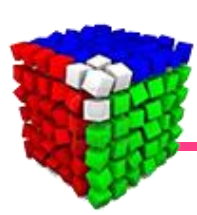




Munsell

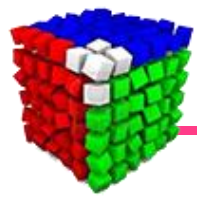
- A Saturação (*Chroma*) é o grau de distância de uma cor qualquer a cor neutra localizada no centro do círculo
- O Brilho (*Value*), representado pelo eixo perpendicular ao círculo, indica a intensidade de luz refletida de uma cor





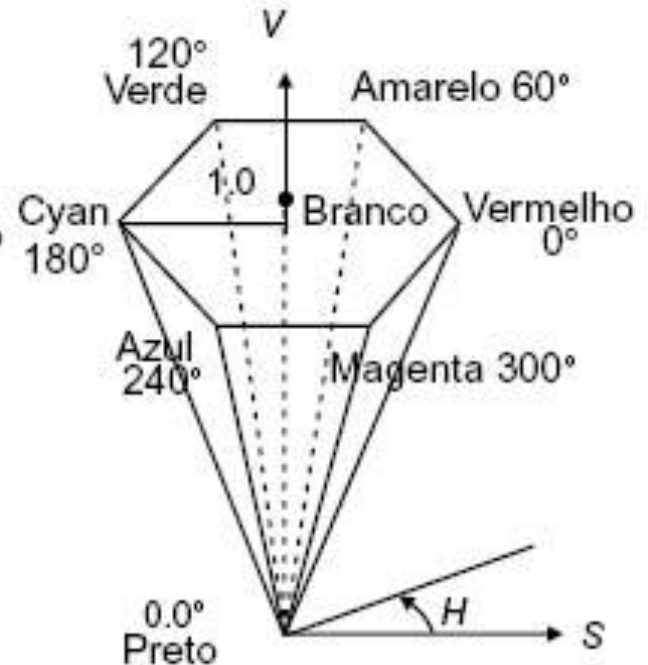
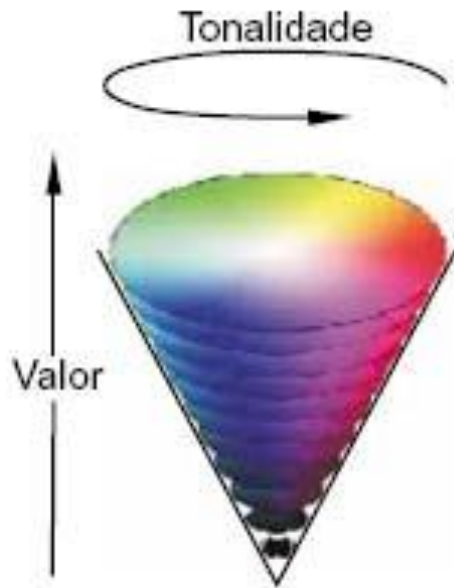
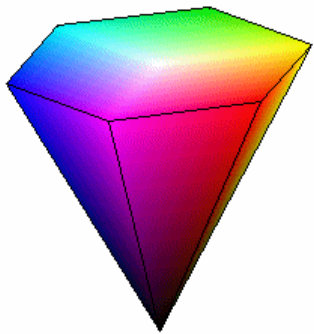
Modelo HSV

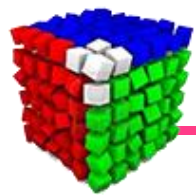
- HSV:
 - *Hue* (Matiz)
 - *Saturation* (Saturação)
 - *Value* (Luminância)
- Desenvolvido em 1978 por Alvey Ray Smith
- Orientado ao usuário - baseado na maneira com que o artista descreve e mistura as cores



Modelo HSV

- Representações do modelo





Modelo HSV - Conversões

■ HSV → RGB

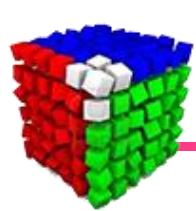
$$H = \begin{cases} 60 \frac{(G - B)}{(M - m)}, & M = R \text{ e } G \geq B \\ 60 \frac{(G - B)}{(M - m)} + 360, & M = R \text{ e } G < B \\ 60 \frac{(B - R)}{(M - m)} + 120, & M = G \\ 60 \frac{(R - G)}{(M - m)} + 240, & M = B \end{cases}$$

$$S = \begin{cases} \frac{(M - m)}{M}, & M \neq 0 \\ 0, & M = 0 \end{cases}$$

$$V = M$$

$$m = \min(R, G, B)$$

$$M = \max(R, G, B)$$



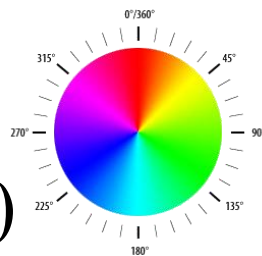
Modelo HSL

- HLS:

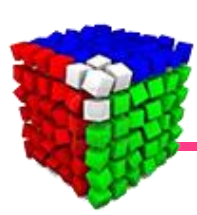
- *Hue* (Matiz)

- *Lightness* (Brilho)

- *Saturation* (Saturação)



- Modelo alternativo para o HSV
- Desenvolvido por Gerald Murch, da Tektronix
- Fácil de ser utilizado
- Definido por um hexágono duplo



Modelo HSL

- HLS

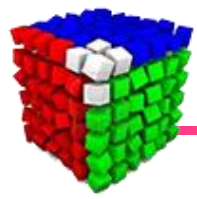
$$L = \frac{M + m}{2}$$

H calculado como antes.

$$S = \begin{cases} \frac{(M - m)}{(M + m)}, & 0 < L \leq 0.5 \\ \frac{(M - m)}{2 - (M + m)}, & L > 0.5 \end{cases}$$

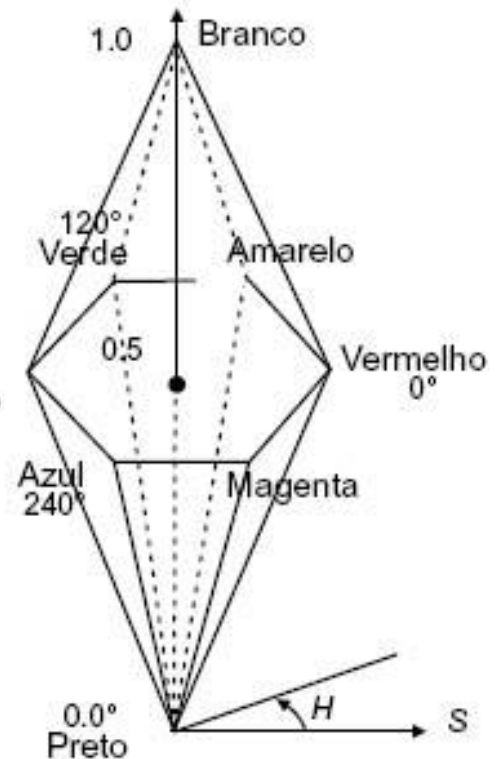
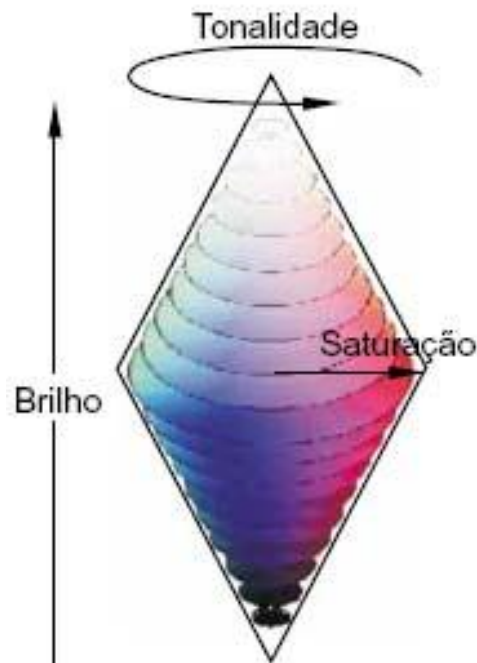
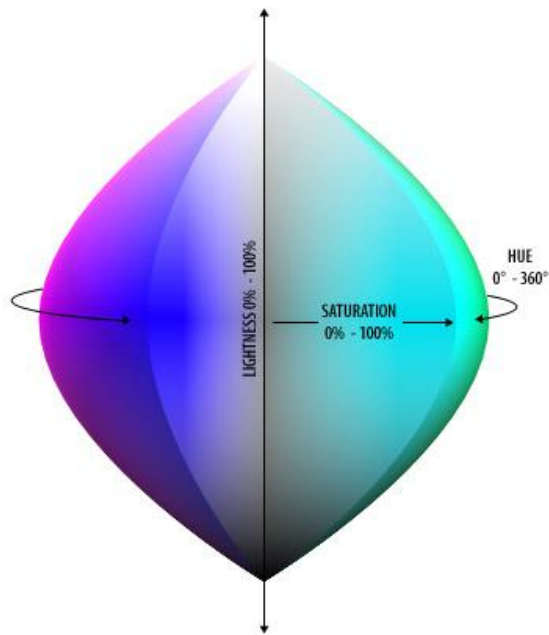
$$m = \min(R, G, B)$$

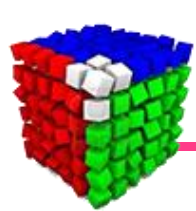
$$M = \max(R, G, B)$$



Modelo HSL

- Representações do modelo





Modelo HSI

- HSI

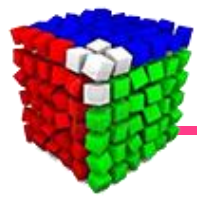
- *Hue* (Matiz)
- *Saturation* (Saturação)
- *Intensity* (Intensidade)

$$H = \begin{cases} \cos^{-1}(\alpha), & B \leq G \\ 360^\circ - \cos^{-1}(\alpha), & B > G \end{cases}$$

$$S = 1 - \frac{3 \min(R, G, B)}{R + G + B}$$

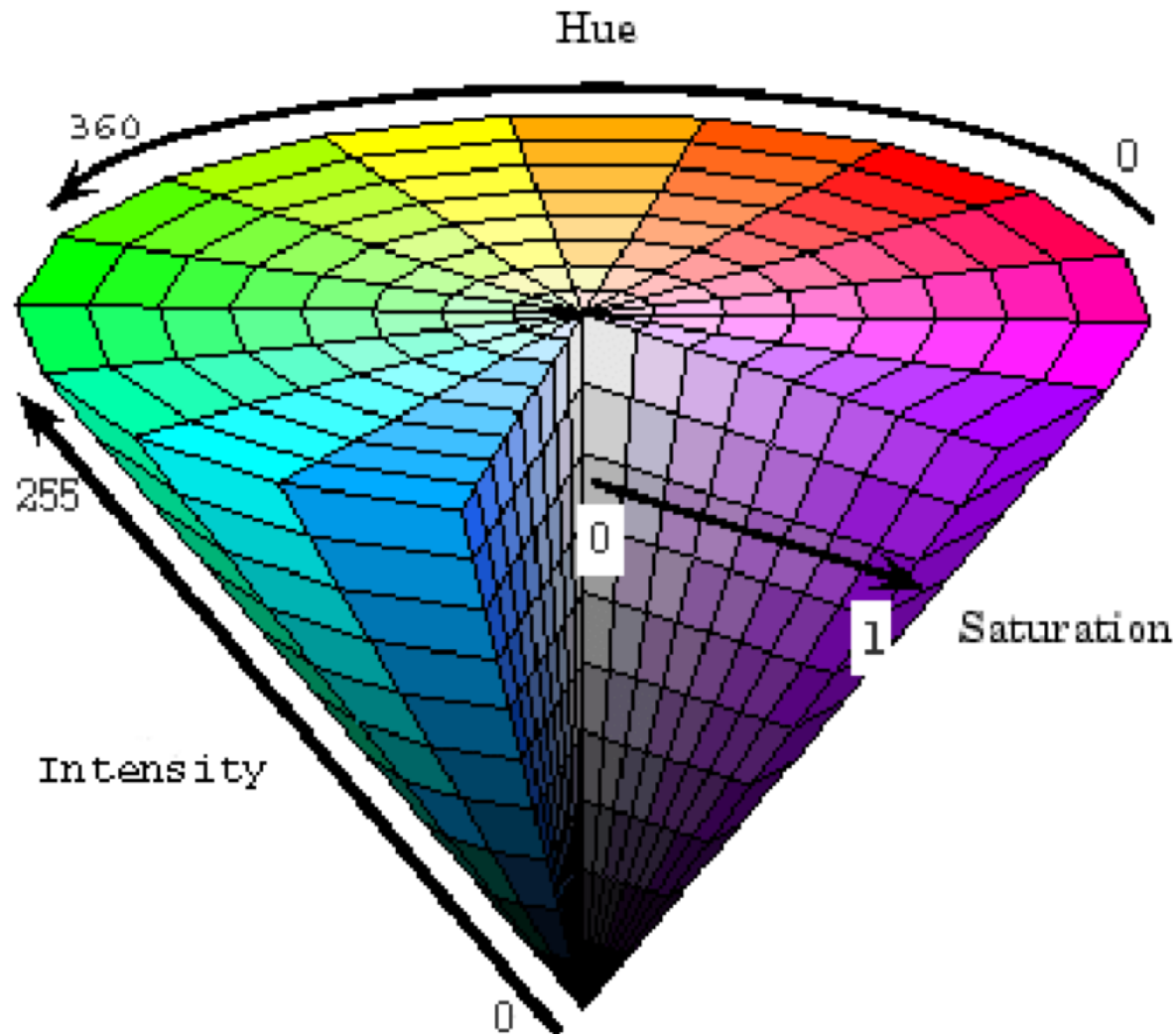
$$I = \frac{R + G + B}{3}$$

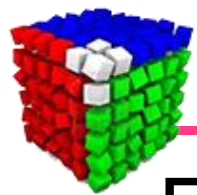
$$\alpha = \frac{(R-G) + (R-B)}{2\sqrt{(R-G)^2 + (R-B)(G-B)}}$$



Modelo HSI

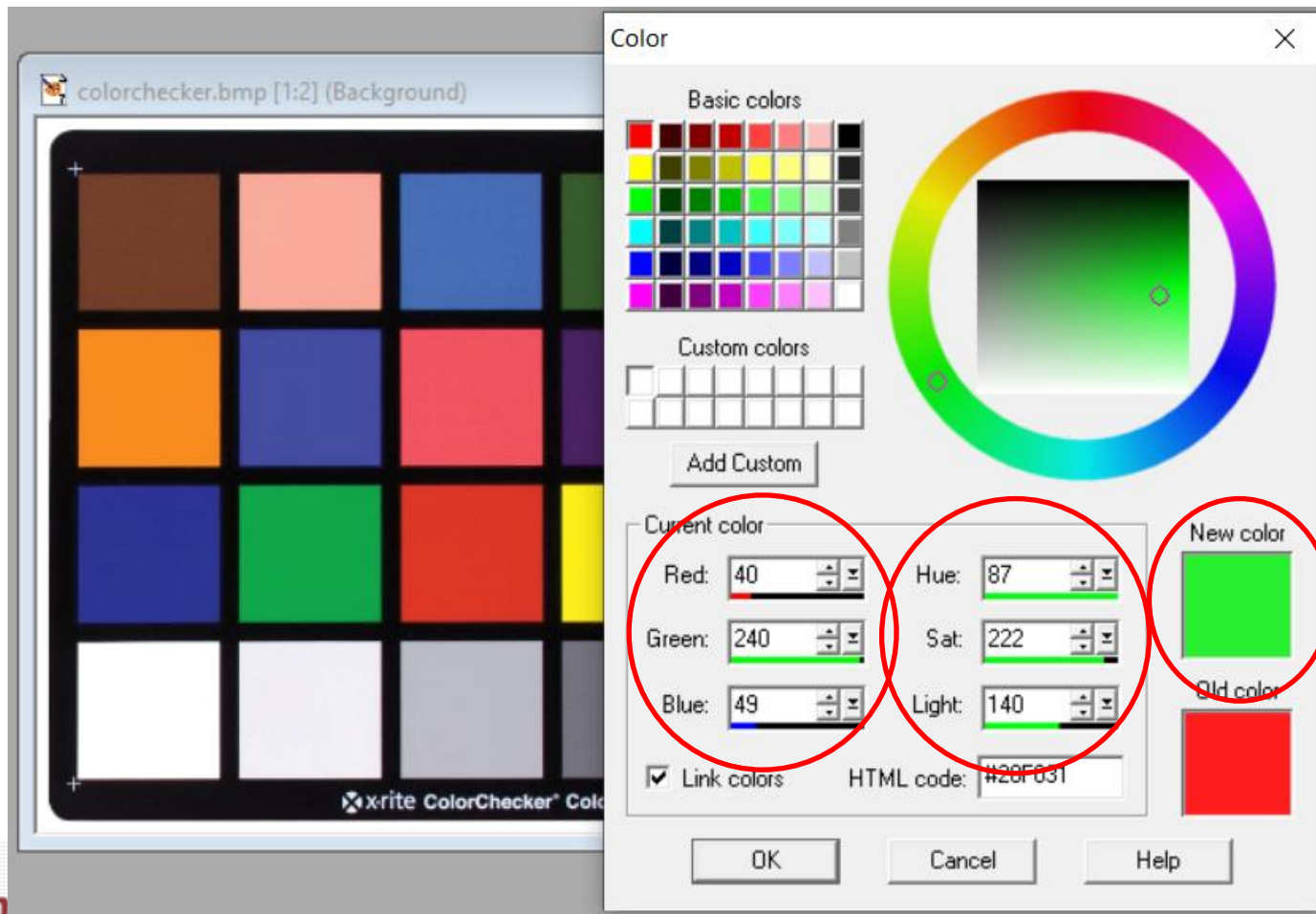
- HSI

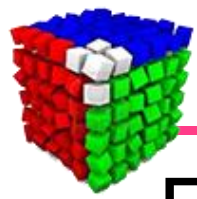




No Computador....

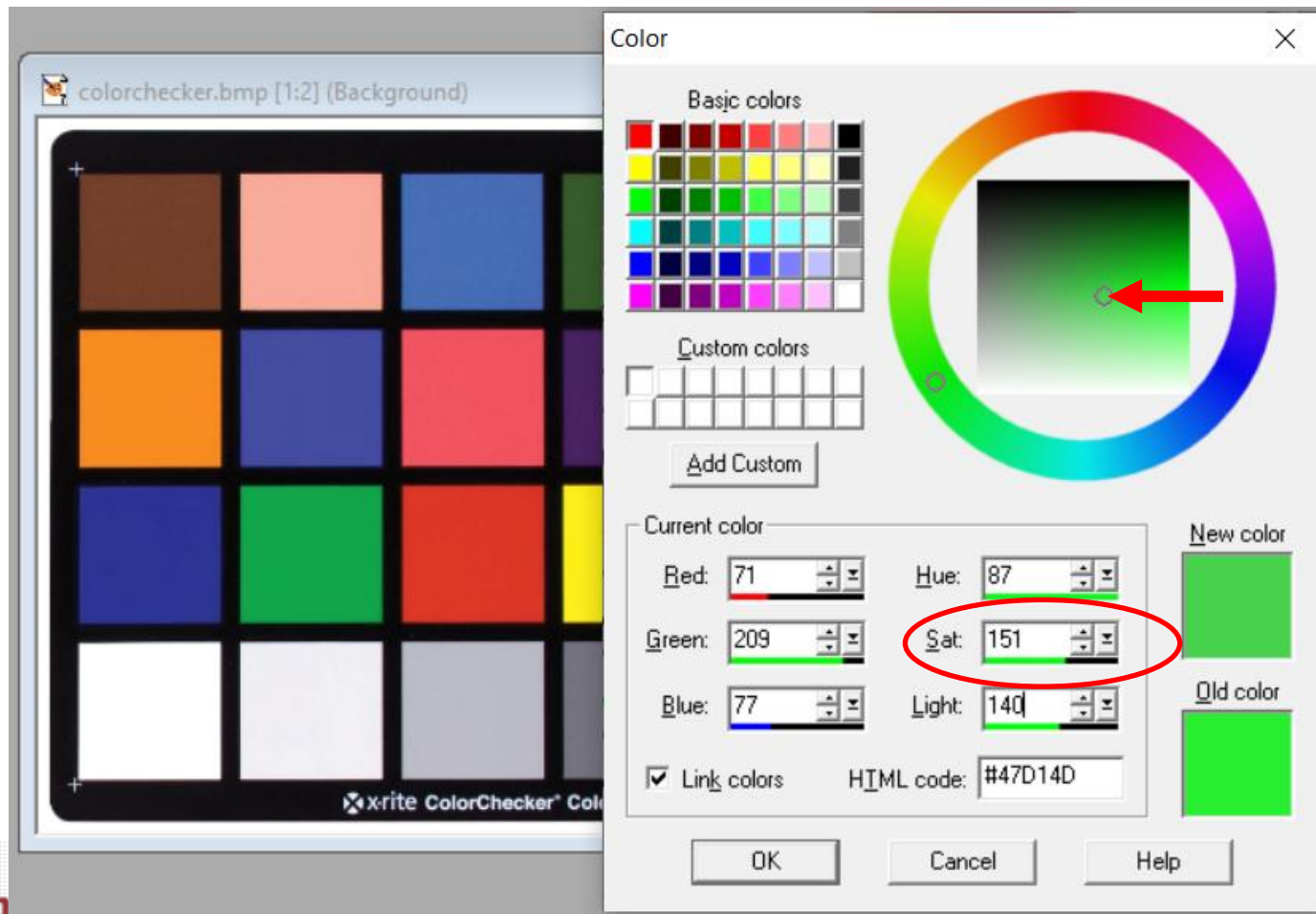
■ Exemplo:

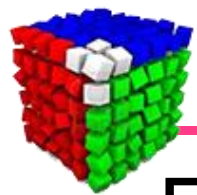




No Computador....

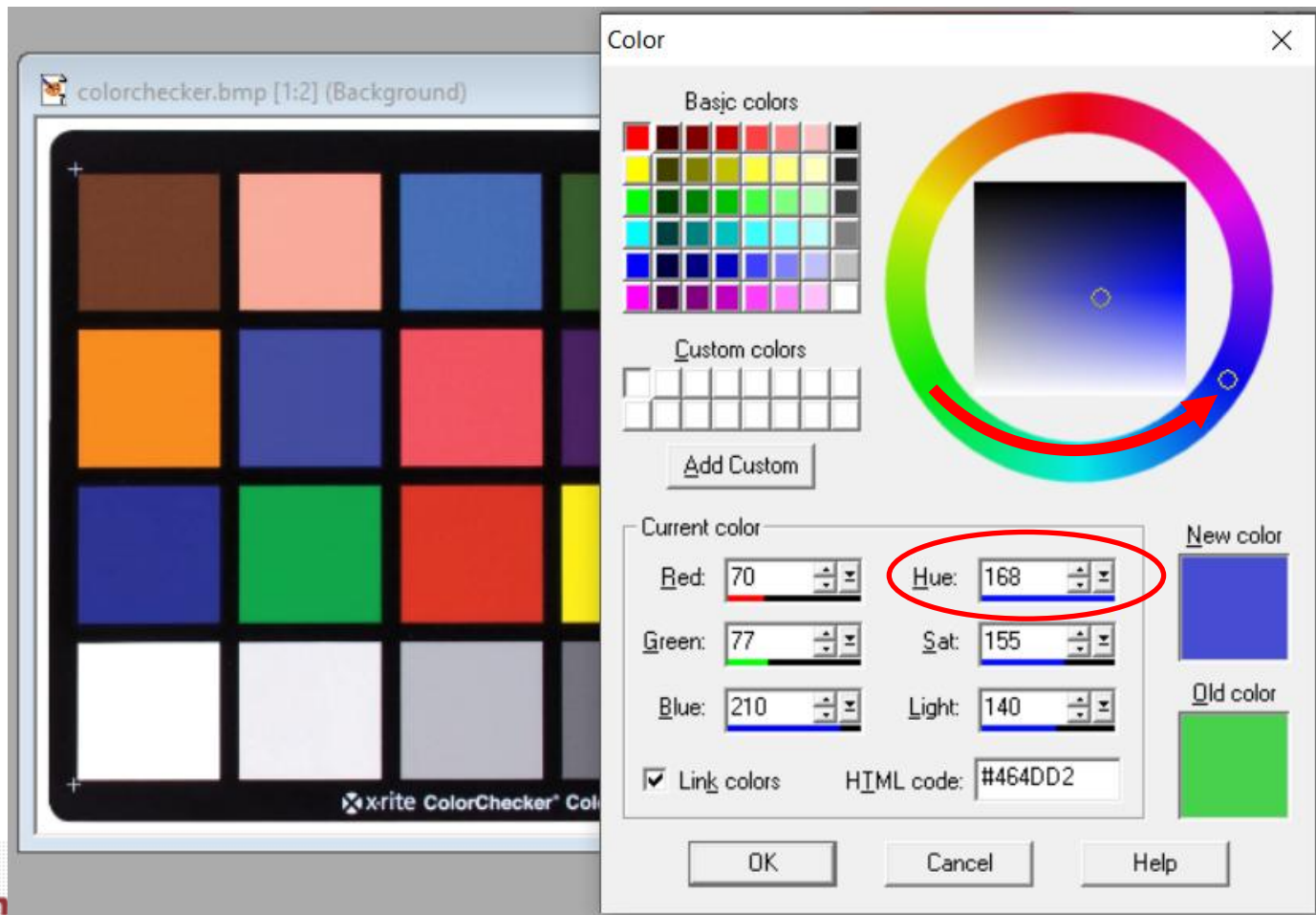
■ Exemplo:

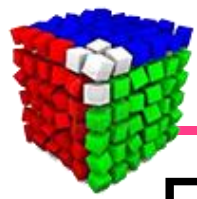




No Computador....

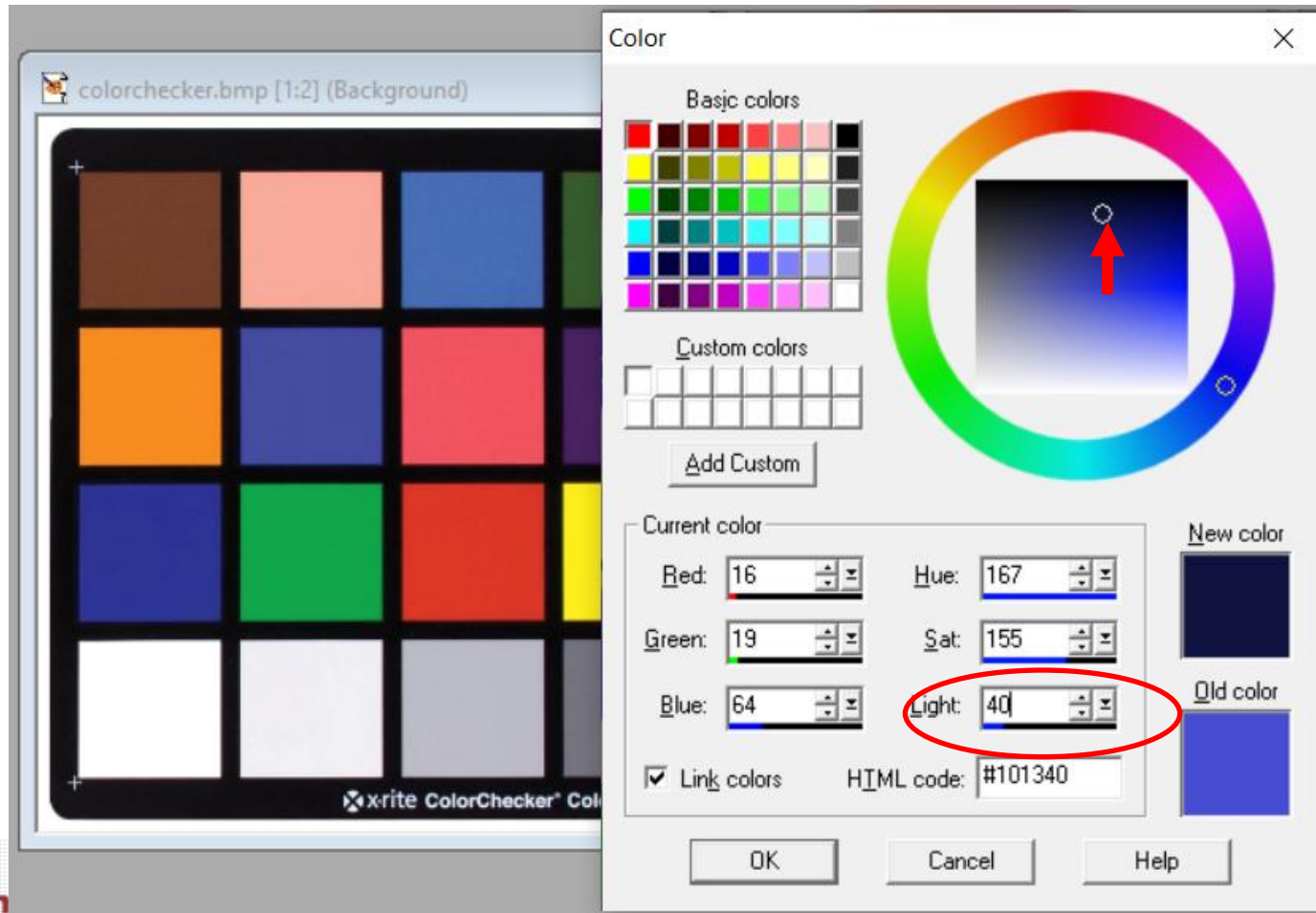
■ Exemplo:





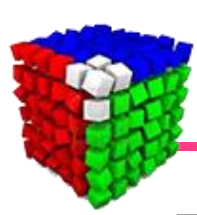
No Computador....

■ Exemplo:



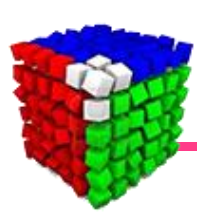
Modelos Uniformes





Modelos Uniformes

- Em 1976, o Comitê Internacional de Iluminação criou dois modelos de cores ditos *uniformes*, ou seja, a distância euclidiana entre duas cores nesses modelos tendem a ter a mesma diferença perceptual
- Nomeados de $L^*u^*v^*$ e $L^*a^*b^*$, mais conhecidos como CIE-LUV e CIE-LAB (ou Luv e Lab), esses modelos são derivados do modelo padrão CIE-XYZ e são bastante utilizados em aplicações de processamento de cor, principalmente quando cores similares são analisadas



CIE L*u*v*

- Modelo uniforme para misturas aditivas de cores

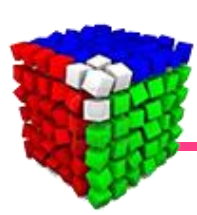
$$L^* = \begin{cases} 116 * \sqrt[3]{\frac{Y}{Y_n}} - 16, & \frac{Y}{Y_n} > 0,008856 \\ 903,3 * \frac{Y}{Y_n}, & \frac{Y}{Y_n} \leq 0,008856 \end{cases}$$

$$u^* = 13 * L^*(u' - u'_n)$$

$$v^* = 13 * L^*(v' - v'_n)$$

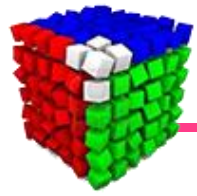
$$u' = 4X/(X + 15Y + 3Z) \quad u'_n = 4X_n/(X_n + 15Y_n + 3Z_n)$$

$$v' = 9Y/(X + 15Y + 3Z) \quad v'_n = 9Y_n/(X_n + 15Y_n + 3Z_n)$$



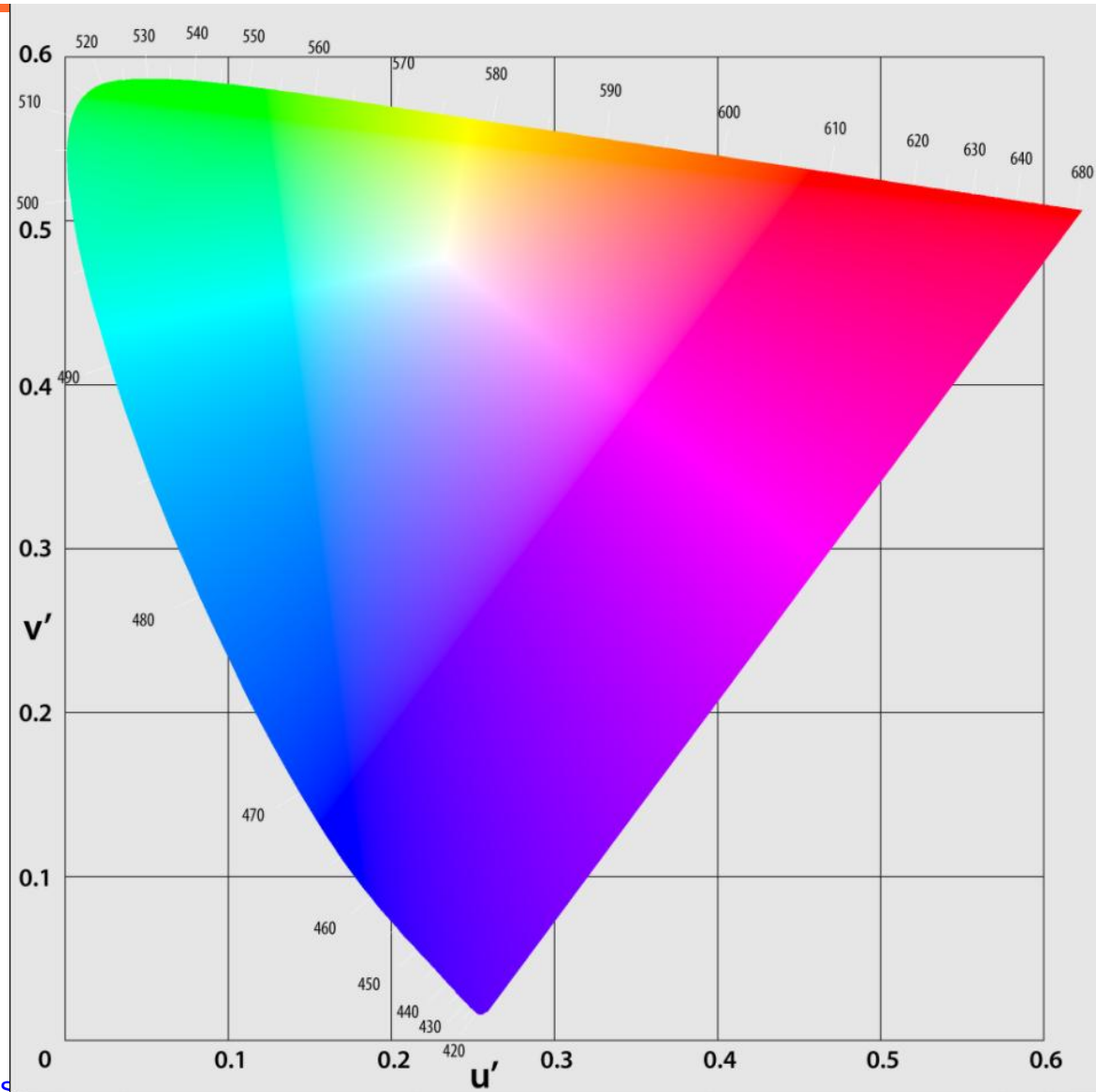
CIE $L^*u^*v^*$

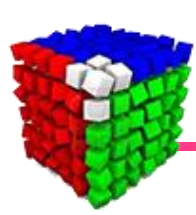
- u^* e v^* estão no intervalo $\pm 100\%$
- $0 \leq L^* \leq 100\%$
- (X_n, Y_n, Z_n) representa a cor branca de referência



CIE $L^*u^*v^*$

Diagrama aproximado,
afinal, tem que ser
apresentado em RGB





CIE L*a*b*

- Modelo uniforme para misturas subtrativas de cores

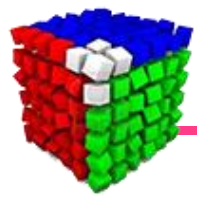
$$L^* = \begin{cases} 116 * \sqrt[3]{\frac{Y}{Y_n}} - 16, & \frac{Y}{Y_n} > 0,008856 \\ 903,3 * \frac{Y}{Y_n}, & \frac{Y}{Y_n} \leq 0,008856 \end{cases}$$

$$a^* = 500 \left[f\left(\frac{X}{X_n}\right) - f\left(\frac{Y}{Y_n}\right) \right]$$

$$b^* = 200 \left[f\left(\frac{Y}{Y_n}\right) - f\left(\frac{Z}{Z_n}\right) \right]$$

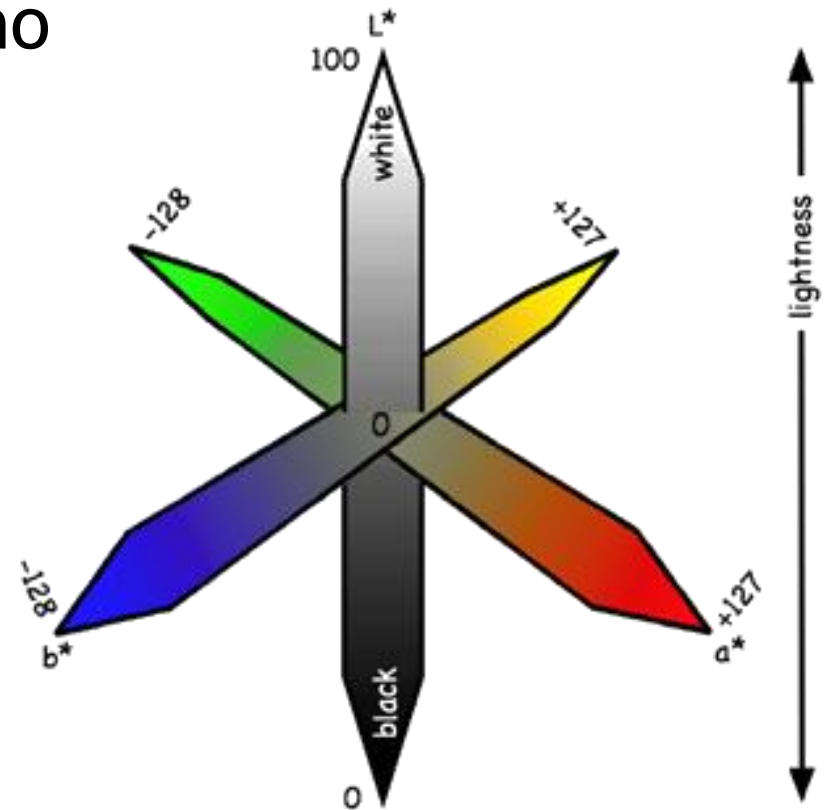
$$f(t) = \begin{cases} \sqrt[3]{t}, & t > 0,008856 \\ 7,787t + \frac{16}{116}, & t \leq 0,008856 \end{cases}$$

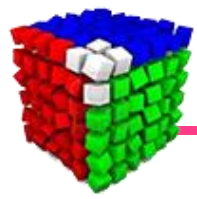
Como antes, (X_n, Y_n, Z_n) é a cor branca de referência.



CIE L*a*b*

- Canal L*: luminância
- Canal a*: verde e vermelho
- canal b*: azul e amarelo

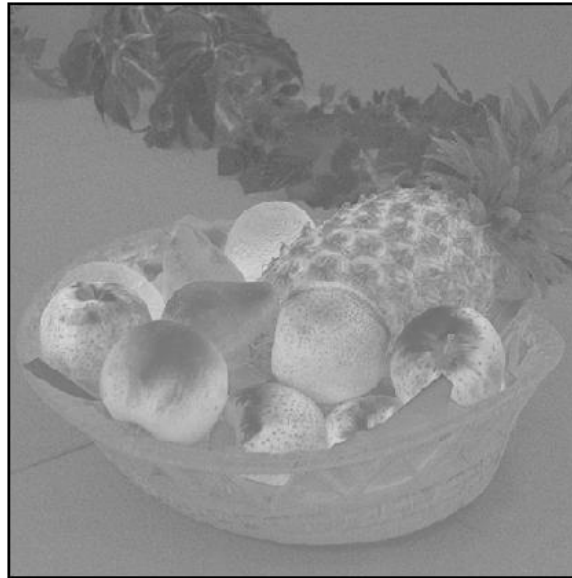




CIE $L^*a^*b^*$



L^*

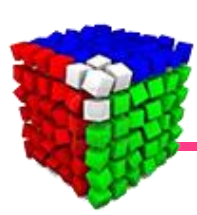


a^*



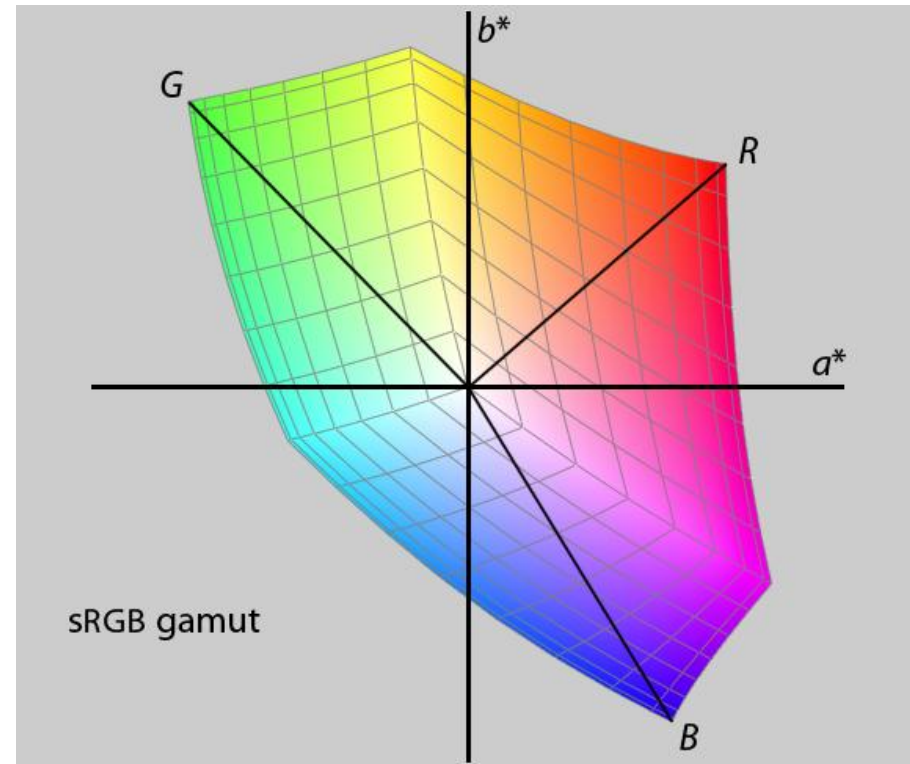
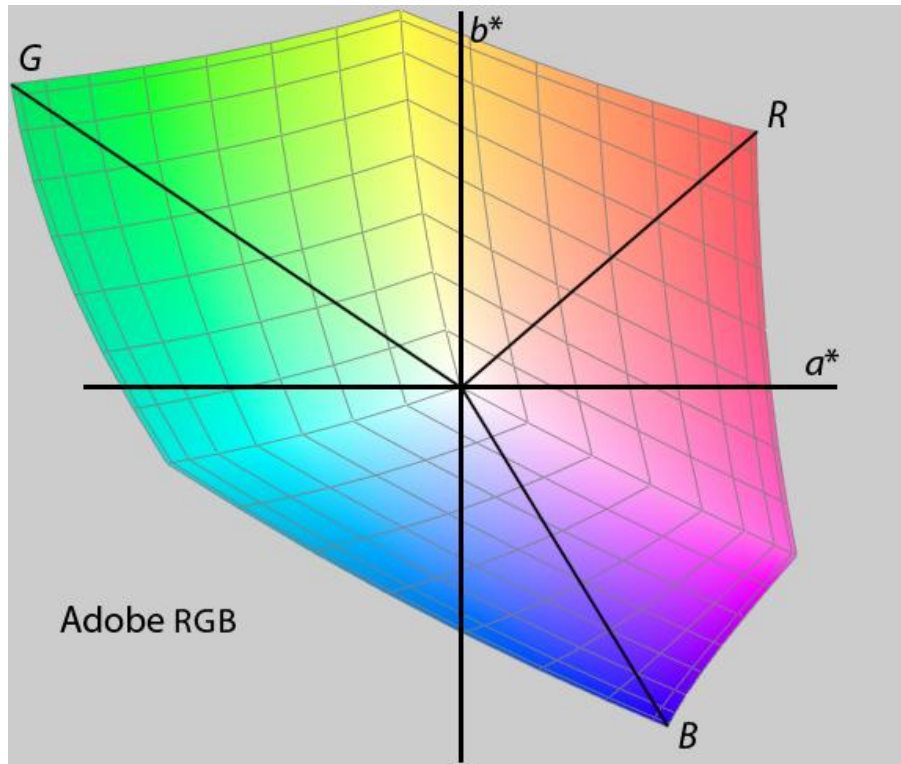
b^*

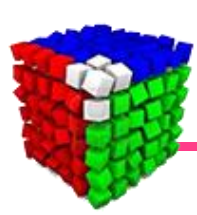
- Obs.: contraste de a^* e b^* aumentado de 40%



CIE $L^*a^*b^*$

- Visualização RGB no $L^*a^*b^*$





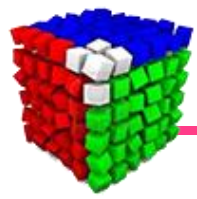
Diferença entre Cores

Distância Euclidiana:

$$\begin{aligned}\text{ColorDist}_{\text{Lab}}(\mathbf{C}_1, \mathbf{C}_2) &= \|\mathbf{C}_1 - \mathbf{C}_2\| \\ &= \sqrt{(L_1^* - L_2^*)^2 + (a_1^* - a_2^*)^2 + (b_1^* - b_2^*)^2}\end{aligned}$$

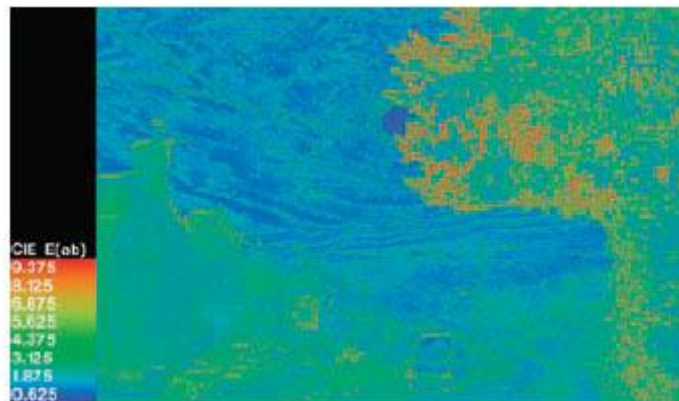
$$\mathbf{C}_1 = (L_1^*, a_1^*, b_1^*)$$

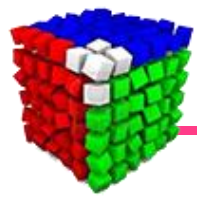
$$\mathbf{C}_2 = (L_2^*, a_2^*, b_2^*)$$



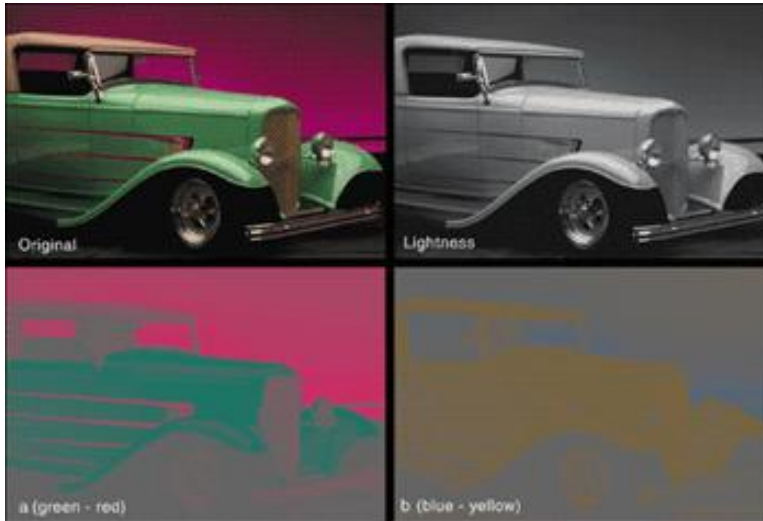
Diferença entre Cores

- Exemplo da utilização da distância - Compressão de imagens





Usando o CIE Lab



Canais Lab

Imagem original



Inversão do canal **a**

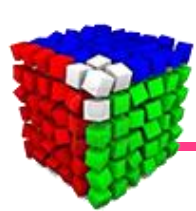


Inversão do canal **b**



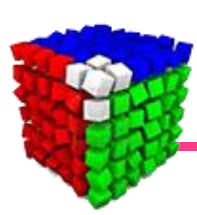
Modelos Baseados em Diferenças Cromáticas





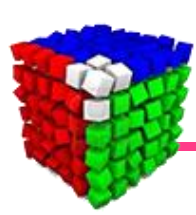
Diferenças Cromáticas

- Os modelos aqui apresentados têm suas transformações fundamentadas em operações de subtração entre as componentes do modelo RGB
 - Podendo, claro, serem utilizados outros modelos como base



Modelo $I_1 I_2 I_3$

- A ideia do modelo é que a operação de subtração entre as componentes do RGB seja responsável por anular os efeitos de reflexão provenientes do tipo da superfície de um objeto
- Por sua vez, a normalização de cada uma dessas subtrações é responsável por anular os efeitos de reflexão dados pelo formato do objeto

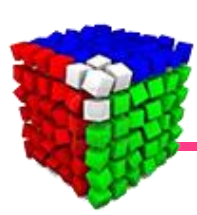


Modelo $l_1l_2l_3$

$$l_1 = \frac{(R - G)^2}{(R - G)^2 + (R - B)^2 + (G - B)^2}$$

$$l_2 = \frac{(R - B)^2}{(R - G)^2 + (R - B)^2 + (G - B)^2}$$

$$l_3 = \frac{(G - B)^2}{(R - G)^2 + (R - B)^2 + (G - B)^2}$$



Modelo $h_1h_2h_3$

- Motivado por três questões:
 - Nenhum uso de operações de divisão e multiplicação para diminuir o custo
 - Cores próximas das cores acromáticas ($R = G = B$) são suprimidas no modelo, salientando as diferenças
 - Anular os efeitos de reflexão na superfície de objetos

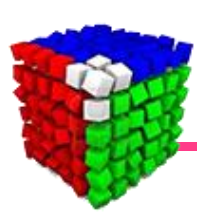
$$h_1 = R - G$$

$$h_2 = G - B$$

$$h_3 = B - R$$

Outros Modelos



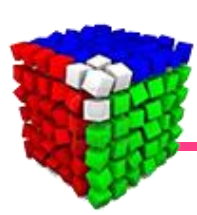


Modelo $I_1 I_2' I_3'$

$$I_1 = \frac{R + G + B}{3}$$

$$I_2' = \frac{R - B}{2}$$

$$I_3' = \frac{2G - R - B}{3}$$

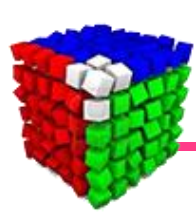


Modelo YO_1O_2

$$Y = 0,2857R + 0,5714G + 0,1429B$$

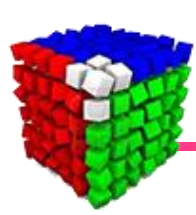
$$O_1 = R - G$$

$$O_2 = 2B - R - G$$



Modelo $R_h G_h B_h$

- Combinação não-linear entre os modelo de cor RGB e $h_1 h_2 h_3$ com a intenção de compor um modelo de cores com informações de cores acromáticas e de diferenças cromáticas



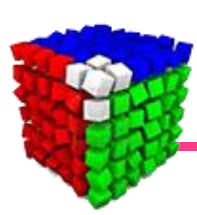
Modelo $R_h G_h B_h$

$$R_h = \alpha \left(\frac{R}{3} \right) + 2 * \max(0, R - G)$$

$$G_h = \alpha \left(\frac{G}{3} \right) + 2 * \max(0, G - B)$$

$$B_h = \alpha \left(\frac{B}{3} \right) + 2 * \max(0, B - R)$$

$$\alpha = \min(R, G, B) / \max(R, G, B)$$

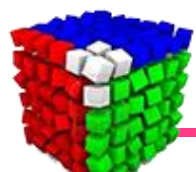


Modelo LgC

- Duas componentes apenas
 - Componente acromática L
 - Unificação das componentes de cromaticidade

$$L = 0,299R + 0,587G + 0,114B$$

$$gC = \sqrt{(R - G)^2 + (G - B)^2 + (B - R)^2}$$



Pantone

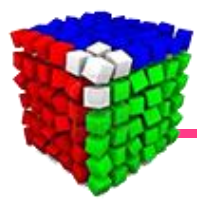
The table below represents the data from the Pantone color chart, showing color names and their corresponding numerical values (Pantone numbers).

Color Name	Pantone Number
Yellow	100
Yellow	110
Yellow	120
Yellow	130
Yellow	140
Yellow	150
Yellow	160
Yellow	170
Yellow	180
Yellow	190
Yellow	200
Yellow	210
Yellow	220
Yellow	230
Yellow	240
Yellow	250
Yellow	260
Yellow	270
Yellow	280
Yellow	290
Yellow	300
Yellow	310
Yellow	320
Yellow	330
Yellow	340
Yellow	350
Yellow	360
Yellow	370
Yellow	380
Yellow	390
Yellow	400
Yellow	410
Yellow	420
Yellow	430
Yellow	440
Yellow	450
Yellow	460
Yellow	470
Yellow	480
Yellow	490
Yellow	500
Yellow	510
Yellow	520
Yellow	530
Yellow	540
Yellow	550
Yellow	560
Yellow	570
Yellow	580
Yellow	590
Yellow	600
Yellow	610
Yellow	620
Yellow	630
Yellow	640
Yellow	650
Yellow	660
Yellow	670
Yellow	680
Yellow	690
Yellow	700
Yellow	710
Yellow	720
Yellow	730
Yellow	740
Yellow	750
Yellow	760
Yellow	770
Yellow	780
Yellow	790
Yellow	800
Yellow	810
Yellow	820
Yellow	830
Yellow	840
Yellow	850
Yellow	860
Yellow	870
Yellow	880
Yellow	890
Yellow	900
Yellow	910
Yellow	920
Yellow	930
Yellow	940
Yellow	950
Yellow	960
Yellow	970
Yellow	980
Yellow	990
Yellow	1000
Yellow	1010
Yellow	1020
Yellow	1030
Yellow	1040
Yellow	1050
Yellow	1060
Yellow	1070
Yellow	1080
Yellow	1090
Yellow	1100
Yellow	1110
Yellow	1120
Yellow	1130
Yellow	1140
Yellow	1150
Yellow	1160
Yellow	1170
Yellow	1180
Yellow	1190
Yellow	1200
Yellow	1210
Yellow	1220
Yellow	1230
Yellow	1240
Yellow	1250
Yellow	1260
Yellow	1270
Yellow	1280
Yellow	1290
Yellow	1300
Yellow	1310
Yellow	1320
Yellow	1330
Yellow	1340
Yellow	1350
Yellow	1360
Yellow	1370
Yellow	1380
Yellow	1390
Yellow	1400
Yellow	1410
Yellow	1420
Yellow	1430
Yellow	1440
Yellow	1450
Yellow	1460
Yellow	1470
Yellow	1480
Yellow	1490
Yellow	1500
Yellow	1510
Yellow	1520
Yellow	1530
Yellow	1540
Yellow	1550
Yellow	1560
Yellow	1570
Yellow	1580
Yellow	1590
Yellow	1600
Yellow	1610
Yellow	1620
Yellow	1630
Yellow	1640
Yellow	1650
Yellow	1660
Yellow	1670
Yellow	1680
Yellow	1690
Yellow	1700
Yellow	1710
Yellow	1720
Yellow	1730
Yellow	1740
Yellow	1750
Yellow	1760
Yellow	1770
Yellow	1780
Yellow	1790
Yellow	1800
Yellow	1810
Yellow	1820
Yellow	1830
Yellow	1840
Yellow	1850
Yellow	1860
Yellow	1870
Yellow	1880
Yellow	1890
Yellow	1900
Yellow	1910
Yellow	1920
Yellow	1930
Yellow	1940
Yellow	1950
Yellow	1960
Yellow	1970
Yellow	1980
Yellow	1990
Yellow	2000
Yellow	2010
Yellow	2020
Yellow	2030
Yellow	2040
Yellow	2050
Yellow	2060
Yellow	2070
Yellow	2080
Yellow	2090
Yellow	2100
Yellow	2110
Yellow	2120
Yellow	2130
Yellow	2140
Yellow	2150
Yellow	2160
Yellow	2170
Yellow	2180
Yellow	2190
Yellow	2200
Yellow	2210
Yellow	2220
Yellow	2230
Yellow	2240
Yellow	2250
Yellow	2260
Yellow	2270
Yellow	2280
Yellow	2290
Yellow	2300
Yellow	2310
Yellow	2320
Yellow	2330
Yellow	2340
Yellow	2350
Yellow	2360
Yellow	2370
Yellow	2380
Yellow	2390
Yellow	2400
Yellow	2410
Yellow	2420
Yellow	2430
Yellow	2440
Yellow	2450
Yellow	2460
Yellow	2470
Yellow	2480
Yellow	2490
Yellow	2500
Yellow	2510
Yellow	2520
Yellow	2530
Yellow	2540
Yellow	2550
Yellow	2560
Yellow	2570
Yellow	2580
Yellow	2590
Yellow	2600
Yellow	2610
Yellow	2620
Yellow	2630
Yellow	2640
Yellow	2650
Yellow	2660
Yellow	2670
Yellow	2680
Yellow	2690
Yellow	2700
Yellow	2710
Yellow	2720
Yellow	2730
Yellow	2740
Yellow	2750
Yellow	2760
Yellow	2770
Yellow	2780
Yellow	2790
Yellow	2800
Yellow	2810
Yellow	2820
Yellow	2830
Yellow	2840
Yellow	2850
Yellow	2860
Yellow	2870
Yellow	2880
Yellow	2890
Yellow	2900
Yellow	2910
Yellow	2920
Yellow	2930
Yellow	2940
Yellow	2950
Yellow	2960
Yellow	2970
Yellow	2980
Yellow	2990
Yellow	3000

<http://finephoto.com.br/rgb-ou-cmyk/>

Aplicações

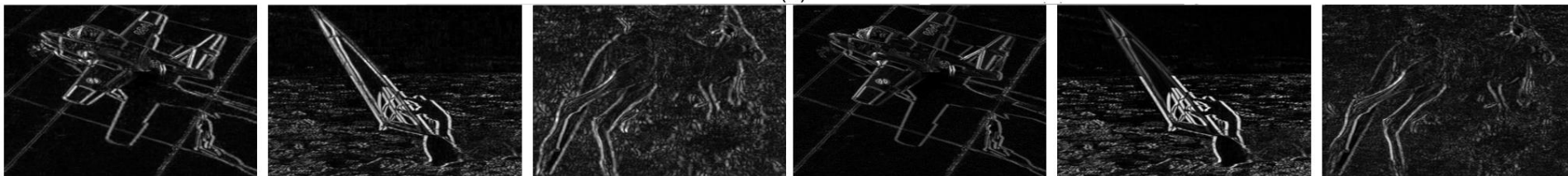




Detecção de Bordas

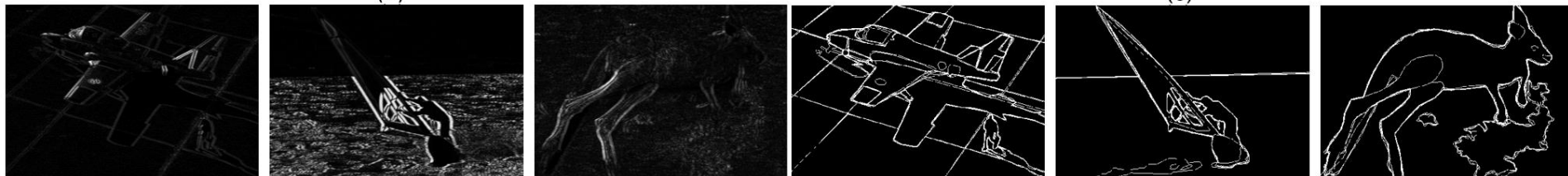


(a)



(b)

(c)



(d)

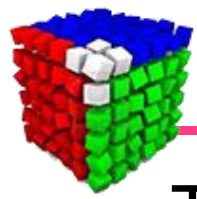
(e)

(a) Imagens originais e bordas detectadas usando os modelos:
 (b) Lab, (c) RGB e (d) XYZ. (e) Padrão ouro.

Sugestão de leitura:

On the Influence of the Color Model for Image Boundary Detection Algorithms based on Convolutional Neural Networks, IEEE IJCNN, 2019

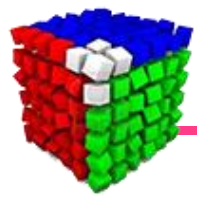
DOI: 10.1109/IJCNN.2019.8851701



Transferência de Cor

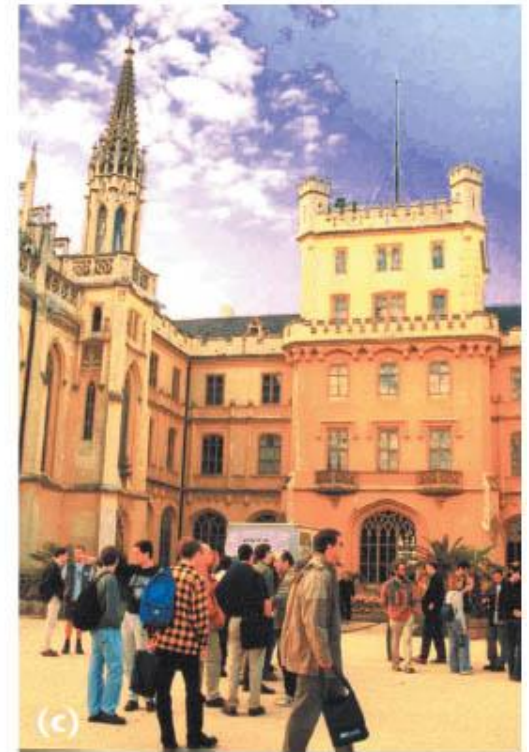
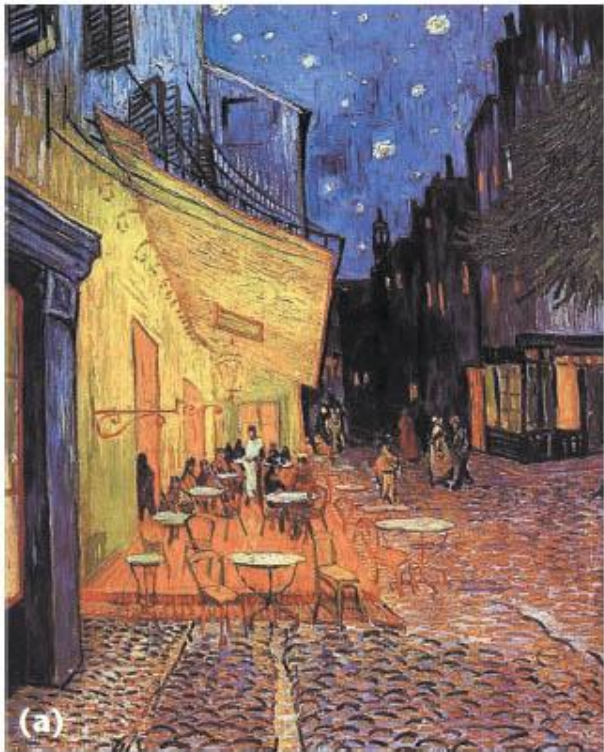
- Transferência de cor entre imagens
 - Proposto por Reinhard *et al.*
 - Utiliza o espaço $l\alpha\beta$
 - Utilizar o canal acromático l para comparação entre imagens
 - Transferir informação cromática dos canais α e β

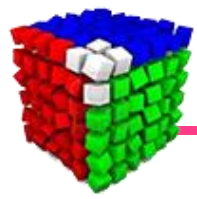




Transferência de Cor

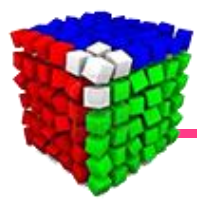
- Outro exemplo





Colorização

- Colorização de imagens em tons de cinza
- Desenvolvido por Welsh *et al.*
- Baseado no trabalho de Reinhard *et al.*

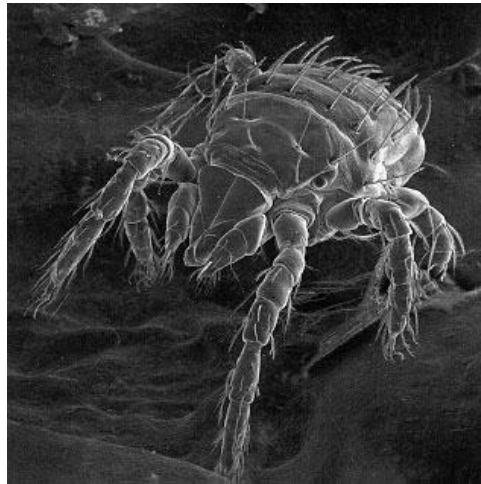


Colorização

- Colorização de imagens em tons de cinza



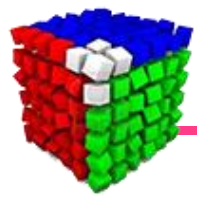
Fonte



Destino



Resultado



Colorização



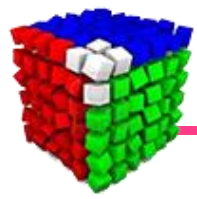
Fonte



Destino



Resultado



Colorização



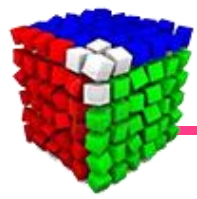
Fonte



Destino



Resultado



Colorização



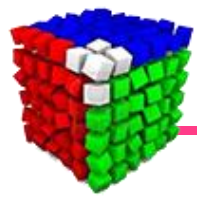
Fonte



Destino



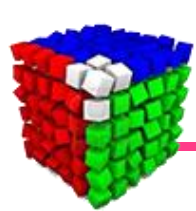
Resultado



Conversão para Tons de Cinza

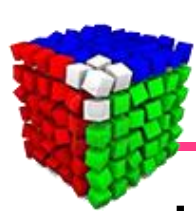
- Conversão para tons de cinza
- Utilizar o canal l (do modelo $l\alpha\beta$)





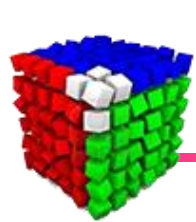
Considerações finais

- Muitos problemas podem ser resolvidos ou simplificados com uma mudança no modelo de cor
- Em geral, isso não acarreta acréscimo no custo computacional
- Normalmente, o custo na conversão de um sistema para outro é $O(1)$



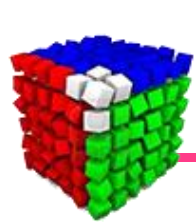
Sugestões de leitura

- Underwater image restoration by color compensation and color-line model
 - Signal Processing: Image Communication, Dez, 2020
 - <https://doi.org/10.1016/j.image.2021.116569>
- QHSL: A quantum hue, saturation, and lightness color model
 - Information Sciences, Out 2021
 - <https://doi.org/10.1016/j.ins.2021.06.077>
- QIRHSI: novel quantum image representation based on HSI color space model
 - Quantum Information Processing, Dez, 2021
 - <https://doi.org/10.1007/s11128-021-03337-0>



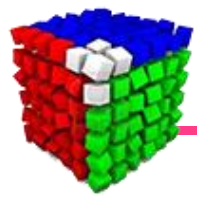
Referências

- J. D. Foley, A. van Dam, S. K. Feiner e J. F. Hughes. “*Computer Graphics: Principles and Practice in C (2nd Edition)*”. Addison-Wesley Professional , 1995.
- H. Pedrini e W. R. Schwartz, “*Análise De Imagens Digitais, Princípios, Algoritmos e Aplicações*”. Thomson Learning, 2007.
- J. Gomes e L. Velho, “*Computação Gráfica: Imagem*”. Instituto de Matemática Pura e Aplicada - IMPA, 1994.
- R. C. Gonzalez e R. E. Woods, “*Digital Image Processing (3rd Edition)*”. Prentice Hall, 2007.
- M. Ebner, “*Color Constancy*”. Wiley, 2007.



Referências

- E. Reinhard, E. A. Khan, A. O. Akyüz, G. M. Johnson, “Color Imaging: Fundamentals and Applications”, A K Peters/CRC Press; Har/Cdr edition, July 22, 2008.
- E. Reinhard, M. Ashikhmin, B. Gooch e P. Shirley, “Color Transfer Between Images”, IEEE Computer Graphics and Applications, vol. 21, no. 5, pp. 34-41, 2001.
- T. Welsh, M. Ashikhmin, K. Mueller, “Transferring Colors to Greyscale Images”, ACM Transactions on Graphics, vol. 21, no. 3, 2002.
- D. Ruderman, D. L. Cronin, T. Welsh and C. C. Chiao, "Statistics of Cone Responses to Natural Images: Implications for Visual Coding". J. Optical Soc. Of America, vol 15, no. 8, 2036-2045, 1998.
- C. Poynton, “*Frequently Asked Questions About Color*”.
http://www.poynton.com/notes/colour_and_gamma/ColorFAQ.html, 2006.
- Munsell - <http://munsell.com/about-munsell-color/>



Próxima aula - Mais sobre Cores

